

Appendix A

PROGRAM LISTING OF THE HEURISTIC COMPILER

This listing includes a rather simple form of the General Compiler [U135], DSCN and SDSC Compilers [U134 and U140, respectively], routines for assembling flow diagrams [U133], and routines for printing JDEF [U125], DSCN [U126], SDSC [U127], and all three of the above [U128]. These routines are called in by a general executive [T1].

List structures in regions E, T, and U, and X5 and X7 are routines. The remaining list structures in region X are data. The version of T1 that is listed will print X102, then compile and print X105 and X100, compile X182 from its flow chart, printing the result, and reconstruct the flow chart of X102 from its code.

The program is in IPL-V (Information Processing Language-V) and is machine independent.

	2	A0		0200	
	2	E0		0200	
	2	L0		0200	
	2	N0		0200	
	2	T0		0200	
	2	U0		0200	
	2	X0		0200	
	5				
E4. PRINT LIST (0) OF DATA TERMS WITHOUT NAMES. ENTER LIST NAME	E4	40H0			E4 000
		J156			E4 010
		E54	91		E4 020
	91	J60			E4 030
		7092			E4 040
		12H0			E4 050
		J157	91		E4 060
	92	E53	J8		E4 070
	E8	1090			E8 000
		J101	E50		E8 010
	90	70	91		E8 020
		12H0			E8 030
		J156			E8 040
		7092			E8 050
		30H0			E8 060
		E54			E8 070
		7093	0		E8 080
	91	E50			E8 090
		J156	E53		E8 100
	92	E50			E8 110
		10N10			E8 120
		J161			E8 130
		52H0			E8 140
		J156	E54		E8 150
	93	E50			E8 160
		10N10			E8 170
		J161	J4		E8 180
E40. MARK ROUTINES ON (0) TO TRACE	E40	1090			E40 000
		J100	0		E40 010
	90	J147	J4		E40 020
E41. MARK ROUTINES ON (0) NOT TO TRACE	E41	1090			E41 000
		J100	0		E41 010
	90	J149	J4		E41 020
PRINT, CLEAR	E50	J155	J154		E50 000
SPACE BAR ONCE	E51	10N1	J161		E51 000
SPACE BAR TWICE	E52	10N2	J161		E52 000
PERIOD, SPACE TWICE	E53	1091			E53 000
		J157	E52		E53 010
	91	+21.			E53 020
COMMA, SPACE ONCE	E54	1091			E54 000
		J157	E51		E54 010
	91	+21,			E54 020
E55. PRINT LIST STRUCTURE, SAVING FOR NEXT PROCESS	E55	E50			E55 000
		40H0	E8		E55 010
	T1	3J0			T1 000
		10A99			T1 010
		J154			T1 020

		13A0		
		10930		
		J100		
		10X102	T1	
		U128	T1	
		10X105	T1	030
		40H0	T1	040
		U135	T1	050
		30H0	T1	060
		U128	T1	070
		10X100	T1	080
		40H0	T1	090
		U135	T1	100
		30H0	T1	110
		U128	T1	120
		10X182	T1	130
		10X41	T1	140
		J10	T1	150
		10X180	T1	160
		10X181	T1	170
		U133	T1	180
		J50	T1	190
		J90	T1	200
		40H0	T1	210
		11W0	T1	220
		10X22	T1	230
		J11	T1	240
		40H0	T1	250
		U128	T1	260
		40H0	T1	270
		U139	T1	280
		40H0	T1	290
		10X26	T1	300
		J10	T1	310
		40H0	T1	320
		U125	T1	330
	930	J147	0	
	T99	J166	J165	
CONSTRUCT IPLV WORD (3), WITH	U100	J52		T99 000
		J90		U100 000
		J136		U100 005
P=(0),Q=(1),SYMB=(2). NO OUTPUT		40H0		U100 006
		11W0		U100 010
ASSIGN P		10X43		U100 020
		J11		U100 030
		40H0		U100 040
		11W1		U100 050
ASSIGN Q		10X44		U100 060
		J11		U100 070
		40H0		U100 080
		11W2		U100 085
ASSIGN SYMB		10X45		U100 090
		J11	J32	U100 100
CONSTRUCT IPLV SYMBOL, WITH	U101	J51		U100 110
				U101 000

REGION=(0),LOCATION=(1). OUTPUT		J90		U101 005
(0) IS LOCAL NAME OF SYMBOL		J136		U101 010
		40H0		U101 020
		11W0		U101 030
ASSIGN REGION		10X33		U101 040
		J11		U101 050
SAVE LOCAL NAME		40H0		U101 055
ASSIGN LOCATION		11W1		U101 060
		10X34		U101 070
		J11	J31	U101 080
CONSTRUCT DESIGNATED SYMBOL,	U102	J51		U102 000
(0)=Q, (1)=SYMB. OUTPUT (0) IS		J90		U102 010
LOCAL NAME OF D.SYMB		J136		U102 020
		40H0		U102 030
		11W0		U102 040
ASSIGN Q VALUE		10X44		U102 050
		J11		U102 060
		40H0		U102 070
		11W1		U102 080
ASSIGN SYMB VALUE		10X45		U102 090
		J11	J31	U102 100
CONSTRUCT - INPUT D.S.,(0) IS D.S.	U103	10N1		U103 000
NO OUTPUT		10X43	J11	U103 010
CONSTRUCT - PROCESS,P=(0),	U104	J50		U104 000
DES.SYMB.=(1). OUTPUT (0) IS LOCAL		40H0		U104 010
NAME OF PROCESS		11W0		U104 020
		10X43		U104 030
		J11	J30	U104 040
CONSTR. J-DEF.,NAME (0),OF (1).	U105	10X41	J11	U105 000
CONSTRUCT PROCESS, P=0, Q=0,	U106	J50		U106 000
(0)=SYMB. OUTPUT (0) IS LOCAL		J90		U106 010
NAME OF PROCESS		J136		U106 020
CREATE WORD		40H0		U106 030
AND MAKE (0) ITS SYMB		11W0		U106 040
		10X45		U106 050
		J11	J30	U106 060
COMPARE IPL SYMBOLS (0) AND (1)	U107	J51		U107 000
FOR IDENTITY. SET H5		11W1		U107 010
		10X33		U107 020
		J10		U107 030
		11W0		U107 040
		10X33		U107 050
		J10		U107 060
TEST IF REGIONS EQUAL.		J114		U107 070
		70J31		U107 080
IF SO,		11W1		U107 090
		10X34		U107 100
		J10		U107 110
		11W0		U107 120
		10X34		U107 130
		J10		U107 140
TEST IF LOCATIONS EQUAL.		J114	J31	U107 150
REPLACE ARGS. IN ROUTINE (0)	U108	J40		U108 000
WITH COPIES.		U114		U108 010

FIND NEXT ARG.	90	J60		U108 020
IF NONE, EXIT		7091		U108 030
		60W0		U108 040
COPY ARGUMENT, AND		52H0		U108 050
		J74		U108 060
INSERT COPY IH LIST		61W0		U108 070
		40H0		U108 080
		10X32		U108 090
		J10		U108 100
TEST IF ARG. IS DETERMINER		10X37		U108 110
		J2		U108 120
		70	92	U108 130
IF NOT, CONTINUE		30H0		U108 140
WITH NEXT ARGUMENT.	93	11W0	90	U108 150
IF A DETERMINER(RECURSE.	92	U108	93	U108 160
NO MORE ARGS., EXIT	91	30H0	J30	U108 170
FIND ON LIST (0) ROUTINE WITH	U110	J42		U110 000
SAME PROCESS(DSCN) AS (1). SET H5		20W0		U110 010
		U111		U110 020
PUT PROC(DSCN) IN W1		20W1		U110 030
COMPARE WITH LIST ITEMS		11W0		U110 040
		1090		U110 050
		J100		U110 060
SET H5 + IF FOUND(- IF NOT		J5	J32	U110 070
STORE ITEM IN W2	90	60W2		U110 080
AND FIND PROC(DSCN)		U111		U110 090
COMPARE IT WITH 1W1		11W1		U110 100
SET H5 - IF FOUND		J2	J5	U110 110
FIND PROCESS OF DSCN	U111	10X20		U111 000
OF ROUTINE (0).		J10		U111 010
(ASSUME IT EXISTS)		10X30	J10	U111 020
TRANSFER H0-LIST ARGUMENT OF	U112	J42		U112 000
ROUTINE (0) TO WS.		60W0		U112 010
BRING IN N-LIST		10X199		U112 020
COPY IT		J73		U112 030
AND STORE IN W1		20W1		U112 040
FIND LIST OF ARGUMENTS		10X20		U112 050
OF DSCN OF (0)		J10		U112 060
		10X31		U112 070
		J10		U112 080
		90		U112 090
		30H0	J32	U112 100
BRING IN TALLY	92	04J0		U112 110
		12W1		U112 112
		10L23		U112 114
		U101		U112 116
		10N1		U112 120
		10N1		U112 121
CONSTRUCT TRANSFER PROCESS		U100		U112 122
INSERT IN J-DEF OF (0)		11W0		U112 130
		J6		U112 140
		10X22	J12	U112 150
LOCATE NEXT ARGUMENT	90	04J60		U112 170
IF NONE, GO TO 92		700		U112 180

	60W2		U112	190
	52H0		U112	200
	10X32		U112	210
FIND TYPE OF ARGUMENT	J10		U112	220
	10X35		U112	230
TEST IF IN H-REGION	J2		U112	240
IF NOT, LOCATE NEXT ARGUMENT	7091		U112	250
IF SO,	12W2		U112	260
COPY OLD ARGUMENT	J74		U112	270
AND MAKE IT LOCAL	J136		U112	275
	11W2		U112	280
	40H0		U112	290
DELETE OLD ARGUMENT	J68		U112	300
	J6		U112	310
INSERT COPY AS NEW ARGUMENT	J63		U112	320
BRING IN ARGUMENT	12W2		U112	330
	40H0		U112	340
ASSIGN X36 AS NEW TYPE	10X36		U112	350
	10X32		U112	360
	J11		U112	370
CHANGE ARG TO WN	U119		U112	380
	11W1		U112	410
	J60		U112	420
	70J7		U112	430
	20W1		U112	440
	92	91	U112	445
	11W2	90	U112	450
LOC. ARGS. IN ROUTINES (2), (1)	U113		U113	000
CORRESPONDING TO HN, N=(0)			U113	010
IN ROUTINE (1)			U113	020
PUT ARG. LIST OF (1) IN W1			U113	030
	20W1			
	40H0			
	U125			
	U114		U113	040
PUT ARG. LIST OF (2) IN W2	20W2		U113	050
	11W1	90	U113	060
LOCATE NEXT ARG. OF (1)	J60		U113	070
IF NONE , TERMINATE	7091		U113	075
AND SAVE IT	60W1		U113	080
	11W2		U113	090
LOCATE NEXT ARG. OF (2)	J60		U113	100
	20W2		U113	110
	52H0		U113	120
	10X32		U113	130
	J10		U113	140
TEST IF ARG (1) IN H0	10X35		U113	150
	J2		U113	160
IF NOT, TRY NEXT ARG.	7090		U113	170
TEST IF LOC (ARG)= (0)	12W1		U113	182
	10X37		U113	184
FIND ITS DETERMINER	J10		U113	186
	70J7		U113	188
SET UP TALLY IN W3	J90		U113	190
	J124		U113	192

		40W3		U113	194
		20W3		U113	196
COUNT NUMBER OF X150S	93	J60		U113	202
		7092		U113	204
		12H0		U113	206
		10X150		U113	208
		J2		U113	210
		7093		U113	212
		11W3		U113	214
		J125		U113	216
		30H0	93	U113	218
COMPARE NUMBER WITH (0)	92	30H0		U113	220
		11W3		U113	222
		11W0		U113	224
		J114		U113	226
		30W3		U113	228
IF NOT, TRY NEXT ARG.		7090		U113	230
IF SO, INPUT LOCATION (2)		11W2		U113	240
AND (1), AND EXIT.		11W1	J32	U113	250
TERMINATE	91	30H0	J32	U113	260
FIND ARG. LIST OF DSCN	U114	10X20		U114	000
OF ROUTINE (0)		J10		U114	010
(ASSUMES IT EXISTS)		10X31	J10	U114	020
COMPILE ROUTINE (1) FROM	U115	J43		U115	000
ROUTINE (0). DSCN PROCESSES SAME		J74	W2=R(1)	U115	010
COPY R(0) AND BRING IN TALLY		J136		U115	015
(1) HAS NAMED ARGS.,		10X199	W1=R(0)	U115	020
(0) HAS H-REGION ARGS.		J60	W0=TALLY	U115	030
		J22		U115	040
	91	11W2		U115	050
		11W1		U115	060
		12W0		U115	070
LOCATE ARGS. FOR HN		U113		U115	080
IF NONE, TERMINATE		7090		U115	090
DISCARD ARG. LIST OF R(0)		30H0		U115	100
		52H0		U115	130
COPY SUBSTITUTE ARGUMENT		J74		U115	150
AND MAKE COPY LOCAL		J136		U115	160
		10N1		U115	210
		10N1		U115	220
MAKE IPL WORD = UUWN		U100		U115	230
		11W1		U115	240
		J6		U115	250
INSERT WORD IN JDEF OF R(0)		10X22		U115	260
		J12		U115	270
		11W0		U115	280
ADVANCE TALLY AND EXIT		J60		U115	290
		20W0	91	U115	295
TERMINATE	90	11W2		U115	300
		11W1		U115	310
TRANSFER JDEF OF R(0) TO R(1).		10X22		U115	320
		J10		U115	330
COPY JDEF		J74		U115	335
AND MARK COPY LOCAL		J136		U115	336

		10X22		U115 340
		J11		U115 350
		11W1		U115 360
ERASE COPY OF R(0) AND EXIT		J72	J33	U115 370
COMPILE JDEF OF (0) FROM ROUTINE(1).	U116	40H0		U116 000
BOTH HAVE SAME PROCESS IN DSCN		10X20		U116 010
(0) HAS H-REGION ARGS.		J10		U116 020
(1) HAS W-REGION ARGS.		10X31		U116 030
FIND ARGS. OF DSCN OF (0)		J10		U116 040
		10N0		U116 050
		J120		U116 060
CREATE TALLY IN W0		J50		U116 070
TALLY NO. OF H-ARGS.		1090		U116 080
		J100		U116 090
		J6		U116 100
		10X22		U116 110
		J10		U116 120
COPY J-DEF OF (1)		J74		U116 130
		40H0		U116 140
		J61		U116 150
FIND LAST SYMBOL OF J-DEF		52H0		U116 160
		10X46		U116 170
ERASE ITS LINK		J14		U116 180
SAVE J-DEF		40H0		U116 190
MAKE INSTR. J5N		10N49		U116 200
		91		U116 210
INSERT AT FRONT OF JDEF		J64		U116 220
		40H0		U116 230
		10N29		U116 240
MAKE INSTR. J3N		91		U116 250
		40H0		U116 260
ASSIGN NO AS LINK OF INSTRUCTION		92		U116 280
INSERT INSTR. AT END OF JDEF		J65		U116 300
		10X22		U116 310
INSERT JDEF IN ROUT. (0).		J11	J30	U116 320
	90	14X32		U116 330
TEST IF ARGUMENT IS		J10		U116 340
IN H-REGION		10X35		U116 350
		J2		U116 360
IF NOT, CONTINUE		70J4		U116 370
IF SO, TALLY AND CONTINUE		11W0		U116 380
		J125	J8	U116 390
	91	4J0		U116 400
		11W0		U116 410
CONSTRUCT NEW INSTRUCTION		J120		U116 420
		40H0		U116 430
		J110		U116 440
		10L10		U116 450
		U101	U106	U116 460
ASSIGN NO AS LINK OF INSTR.	92	4J90		U116 470
CREATE LOCAL SYMBOL		J136		U116 480
		40H0		U116 490
SAVE IT		J50		U116 500
BRING IN NO		10N0		U116 510

MAKE IT REGION OF SYMBOL		10X33		U116 520
		J11		U116 530
MAKE 1W0 LINK OF INSTR.		11W0		U116 540
		10X46		U116 550
		J11	J30	U116 560
REPLACE HS BY WS IN DSCN	U117	J90		U117 000
OF (0). OUTPUT (0)=N, NUMBER		J124		U117 010
OF REPLACEMENTS.		J50	1W0=TALLY	U117 020
		10X20		U117 030
		J10		U117 040
FIND LIST OF ARGS OF DSCN OF (0)		10X31		U117 050
		J10		U117 060
DO 90 TO ARGS ON LIST		1090		U117 070
		J100		U117 080
BRING IN TALLY AND EXIT		11W0	J30	U117 090
SAVE ARG.	90	44H0		U117 100
		10X32		U117 110
		J10		U117 120
FIND ITS TYPE		10X37		U117 130
		J2		U117 140
TEST IF DETERMINER		70	91	U117 150
		40H0		U117 155
DESIG. SYMB. FIND REGION		10X45		U117 160
		J10		U117 170
		10X33		U117 190
		J10		U117 200
TEST IF H-REGION		10L8		U117 210
		J2		U117 220
IF NOT, EXIT		7092		U117 230
CHANGE ARG TO WN		U119		U117 240
AND ADD ONE TO TALLY		11W0		U117 270
		J125		U117 280
	92	30H0	J4	U117 290
DETERMINER.	91	10X20		U117 300
FIND ITS LIST OF ARGUMENTS		J10		U117 310
		10X31		U117 320
		J10		U117 330
		1090		U117 340
		J100	0	U117 350
MODIFY JDEF OF (0) TO TAKE N=(1)	U118	J51		U118 000
INPUTS FROM WS. NO OUTPUT.		11W0		U118 010
		10X22		U118 020
		J10		U118 030
FIND AND SAVE JDEF OF (0)		40H0		U118 040
FIND ITS LAST SYMBOL		J61		U118 050
		52H0		U118 060
		10X46		U118 070
ERASE ITS LINK		J14		U118 080
SAVE JDEF		40H0		U118 090
MAKE INSTR. J5N		10N49		U118 100
		91		U118 110
INSERT AT FRONT OF JDEF		J64		U118 120
MAKE INSTR. J3N		10N29		U118 140
		91		U118 150

ASSIGN NO AS LINK OF INSTR.		40H0		U118 160
INSERT INSTR. AT END OF JDEF		92		U118 170
CONSTRUCT NEW INSTRUCTION		J65	J31	U118 180
BRING IN N	91	4J0		U118 190
COMPUTE XN		11W1		U118 200
		J120		U118 210
		40H0		U118 220
		J110		U118 230
MAKE JXN		10L10		U118 240
		U101	U106	U118 250
ASSIGN NO AS LINK OF INSTR.	92	4J90		U118 260
CREATE LOCAL SYMBOL		J136		U118 270
		40H0		U118 280
SAVE IT		J50		U118 290
BRING IN NO		10N0		U118 300
		10X33		U118 310
MAKE IT REGION OF SYMBOL		J11		U118 320
MAKE SYMBOL LINK OF INSTR.		11W0		U118 330
		10X46		U118 340
		J11	J30	U118 350
CHANGE ARG (0)=HN TO WN,	U119	40H0		U119 000
NO OUTPUT		10X37		U119 010
FIND ITS DETERMINER		J10		U119 020
SET UP TALLY IN WC		J90		U119 040
		J124		U119 050
		J50		U119 060
TALLY AND ERASE X150S	93	40H0		U119 070
FROM DETERMINER		10X150		U119 080
		J62		U119 090
		7094		U119 100
		11W0		U119 110
		J125		U119 120
		30H0		U119 130
		J68	93	U119 140
	94	30H0		U119 150
CONSTRUCT SYMBOL WN		30H0		U119 160
		11W0		U119 170
		30W0		U119 180
		10L23		U119 190
		U101		U119 200
ASSIGN IT AS X45 OF ARGUMENT		10X45		U119 210
		J11	0	U119 220
MODIFY JDEF TO DO PROCESS (2)	U120	10N9		U120 005
AFTER (3) WITH FIELD (1)=(0). SET		U101		U120 010
H5+ IF (2) WAS NAMELESS, H5- IF		J53		U120 015
(2) HAS NAME. NO OUTPUT.		11W2		U120 020
		J81		U120 025
FIND FIRST INSTR. OF (1)		40H0		U120 030
		10X41		U120 040
IF IT HAS NAME, REPLACE (0)		J10		U120 050
WITH NAME, IF NOT, SKIP TO 90.		7090		U120 060
		20W0		U120 070
		30H0	91	U120 080
PUT (0) AS NAME OF FIRST OF (1).	90	11W0		U120 090

		10X41		U120 100
		J11		U120 110
FIND LAST INSTR. OF (3)	91	11W3		U120 120
		J61		U120 130
		52H0		U120 140
MAKE 1W0 ITS LINK.		11W0		U120 150
OR SYMB		11W1		U120 160
		J11		U120 170
SET H5 AND EXIT		J5	J33	U120 180
MODIFY JDEF TO EXIT AFTER	U122	J61		U122 000
PROCESS (0). NO OUTPUT		52H0		U122 010
		40H0		U122 012
		10X46		U122 014
		J10		U122 016
		70	90	U122 018
MAKE SYMB. 0.		10N0		U122 020
		40H0		U122 030
		U101		U122 040
ASSIGN IT AS LINK OF LAST OF (0).		10X46	J11	U122 050
	90	30H0	J8	U122 060
PRINT THE JDEF OF (0)	U125	10X22		U125 000
FIND JDEF		J10		U125 010
CLEAR PRINT LINE		J154		U125 020
PRINT LIST OF INSTRUCTIONS		1090		U125 030
		J100	E50	U125 040
PROCESS INSTRUCTION	90	04J0		U125 050
		10X40		U125 055
		10N28		U125 060
ENTER TYPE IN COL. 28		920		U125 070
		10X41		U125 080
		10N30		U125 090
ENTER NAME IN COLS. 30-34		910		U125 100
		10X42		U125 110
		10N35		U125 120
ENTER SIGN IN COL. 35		920		U125 130
		10X43		U125 140
		10N36		U125 150
ENTER P IN COL. 36		920		U125 160
		10X44		U125 170
		10N37		U125 180
ENTER Q IN COL. 37		920		U125 190
		10X45		U125 200
		10N38		U125 210
ENTER SYMB IN COLS. 38-42		910		U125 220
		10X46		U125 230
		10N44		U125 240
ENTER LINK IN COLS. 44-48.		910		U125 245
PRINT.		E50		U125 250
		30H0	J4	U125 255
FIND VALUE OF ATTRIBUTE (1)	910	04930		U125 260
AND ENTER ITS REGION AND		70J31		U125 270
LOCATION IN COLS. BEGINNING		J160		U125 280
		40H0		U125 285
AT (0).		10X33		U125 290

ENTER REGION		J10		U125 300
		70912		U125 310
		J157		U125 320
	911	10X34		U125 330
ENTER LOCATION		J10		U125 340
		70J31		U125 350
		J157	J31	U125 360
	912	10N1		U125 365
		J161	911	U125 367
FIND VALUE OF ATTRIBUTE (1)	920	04930		U125 370
AND ENTER IN COL. (0)		70J31		U125 380
		J160		U125 390
		J157	J31	U125 400
PROCESS FOR 910 AND 920	930	04J51		U125 410
		40H0		U125 420
		11W1		U125 430
		J10		U125 440
		700		U125 450
		11W0	0	U125 460
ENTER IN PRINT LINE THE NAME OF	U126	J41		U126 000
		40H0		U126 001
TEST IF ROUTINE		10X32		U126 002
IS A DETERMINER		J10		U126 003
		7096		U126 004
		10X37		U126 005
		J2		U126 006
IF NOT, GO TO 96		7096		U126 007
IF SO, PRINT ITS VERB		40H0		U126 008
		10X20		U126 009
		J10		U126 010
ENTER DATA TERMS OF VERB	98	J60		U126 011
		7097		U126 012
		12H0		U126 013
		910	98	U126 014
	97	E51		U126 015
AND CONTINUE AT 96		30H0	96	U126 016
	96	10X20		U126 019
FIND ITS DSCN		J10		U126 020
		40H0		U126 030
		10X30		U126 040
		J10		U126 050
PUT PROCESS IN W0		20W0		U126 060
		10X31		U126 070
		J10		U126 080
PUT ARGS IN W1		20W1		U126 090
	91	11W0		U126 100
FIND NEXT WORD IN NAME		J60		U126 110
EXIT IF NONE		7092		U126 120
		60W0		U126 130
		52H0		U126 140
		40H0		U126 150
TEST IF 1H0 NAMES DATA TERM		J131		U126 160
		7090		U126 170
		910	91	U126 180

DOES NOT NAME DATA TERM	90	30H0		U126 190
		11W1		U126 191
SPACE ONCE		E51		U126 195
FIND NEXT ARGUMENT		J60		U126 200
ERROR STOP		70J7		U126 210
		60W1		U126 220
		52H0		U126 230
		40H0		U126 240
		10X32		U126 250
		J10		U126 260
		10X37		U126 270
TEST IF ARG IS A DETERMINER		J2		U126 280
		70	94	U126 290
IF NOT, SAVE IT		40H0		U126 300
TEST IF MEMBER OF HO LIST		10X32		U126 301
		J10		U126 302
		10X35		U126 303
		J2		U126 304
IF SO, BRANCH TO 95		70	95	U126 305
IF NOT, SAVE IT		40H0		U126 306
FIND ITS DETERMINER		10X37		U126 310
		J10		U126 315
		1093		U126 320
AND ENTER IT		J100		U126 330
SPACE ONCE		E51		U126 335
FIND ITS SYMBOL		10X45		U126 340
		J10		U126 350
		40H0		U126 360
		10X33		U126 370
		J10		U126 380
		910		U126 390
		10X34		U126 400
		J10		U126 410
		910		U126 420
		E51	91	U126 425
	92	30H0	J31	U126 430
	93	10910		U126 440
		J100	0	U126 450
	94	U126	91	U126 460
ENTER IN PRINT LINE	910	40H0		U126 470
		J157		U126 475
LINE SPACE IF NO ROOM		70	J8	U126 480
		E50		U126 490
		10N10		U126 500
		J160	910	U126 510
ENTER NAME OF MEMBER	95	J60		U126 530
OF HO LIST		70J7		U126 540
		52H0		U126 550
		910		U126 560
		E51	91	U126 570
PRINT THE SDSC OF ROUTINE (0)	U127	J40		U127 000
CLEAR PRINT LINE		J154		U127 010
		10X24		U127 020
FIND SDSC		J10		U127 030

FIND LIST OF AFFECTED CELLS		70J30		U127 040
		10X71		U127 050
		J10		U127 060
DO 90 TO AFFECTED CELLS		70J30		U127 070
AND EXIT		1090		U127 080
		J100	J30	U127 090
	90	60W0		U127 100
FIND NAME OF CELL		10X41		U127 110
		J10		U127 120
ENTER IT IN PRINT LINE		91		U127 130
		E53		U127 140
		11W0		U127 150
FIND INPUT LIST		10X75		U127 160
		J10		U127 170
AND ENTER ITS SYMBOLS		1092		U127 180
		J100		U127 190
ENTER PERIOD		E53		U127 200
		11W0		U127 210
FIND OUTPUT LIST		10X76		U127 220
		J10		U127 230
ENTER ITS SYMBOLS		1092		U127 240
		J100		U127 250
ENTER PERIOD AND PRINT LINE		E53	E50	U127 260
ENTER A SYMBOL IN THE PRINT LINE	91	40H0		U127 270
		10X33		U127 280
		J10		U127 290
		J157		U127 300
		10X34		U127 310
		J10	J157	U127 320
ENTER THE SYMBOLS, THEN COMMAS	92	91	E54	U127 330
PRINT NAME, DSCN, JDEF, AND	U128	J50		U128 000
SDSC OF ROUTINE (0)		90		U128 010
		E50		U128 020
		11W0		U128 040
		40H0		U128 050
		10X20		U128 060
TEST OF IT HAS A DSCN		J10		U128 070
IF NOT, SKIP TO 92		7092		U128 080
		30H0		U128 090
ENTER DSCN IN PRINT LINE		U126		U128 100
PRINT		E50		U128 110
AND SPACE		E50		U128 120
		11W0		U128 130
	92	40H0		U128 140
		10X22		U128 150
TEST IF IT HAS A JDEF		J10		U128 160
IF NOT, SKIP TO 92		7093		U128 170
		30H0		U128 180
PRINT JDEF		U125		U128 190
AND SPACE		E50		U128 200
		11W0		U128 210
	93	40H0		U128 220
		10X24		U128 230
TEST IF IT HAS A SDSC		J10		U128 240

IF NOT, EXIT BY 95		7095		U128 250
		30H0		U128 290
PRINT SDSC AND EXIT		U127	J30	U128 300
	95	30H0	J30	U128 310
	90	04J0		U128 320
		11W0		U128 330
TEST IF IT HAS AN IPLN		10X25		U128 340
		J10		U128 350
IF NOT, EXIT		700		U128 360
		40H0		U128 370
IF SO, ENTER IN PRINT LINE		10X33		U128 380
		J10		U128 390
		J157		U128 400
		10X34		U128 410
		J10	J157	U128 420
COMPILE JDEF OF ROUT.(0) FROM	U130	J43		U130 000
JDEF OF ROUT.(1). ROUT(1) IS A J.		J21		U130 010
		11W0		U130 020
COPY R1W0 AND		J74		U130 030
COPY ARGS OF DSCN		40H0		U130 031
		U108		U130 040
		60W2		U130 043
REPLACE HS BY WS IN DSCN OF COPY.		U117		U130 050
SAVE(0)=NUMBER OF REPLACEMENTS.		20W3		U130 060
		11W1		U130 070
COMPILE JDEF OF 1W2		11W2		U130 080
FROM JDEF OF 1W1.		U131		U130 090
MODIFY JDEF OF 1W2 TO TAKE		11W3		U130 100
N=1W3 INPUTS FROM WS.		11W2		U130 110
		U118		U130 120
BRING IN 1W0.		11W0		U130 130
		11W2		U130 140
FIND JDEF OF 1W2		10X22		U130 150
AND ASSIGN IT AS JDEF OF 1W0		J10		U130 160
		J74		U130 161
		J136		U130 162
		10X22		U130 165
		J11		U130 170
ASSIGN NAME TO JDEF.		910		U130 175
		11W2		U130 180.
ERASE 1W2,		J72		U130 190
AND TALLY CELL,		11W3		U130 200
AND EXIT		J9	J33	U130 210
	910	11W0		U130 220
FIND JDEF		10X22		U130 230
		J10		U130 240
FIND FIRST INSTRUCTION.		J81		U130 250
		40H0		U130 260
		10X41		U130 265
		J10		U130 266
IF IT HAS NO NAME,		70	911	U130 270
		11W0		U130 280
		10X25		U130 290
ASSIGN IT V(X25) OF ROUTINE.		J10		U130 300

IF IT HAS NAME, EXIT.	911	10X41	J11	U130	310
COMPILE JDEF OF ROUT.(0)	U131	30H0	J8	U130	320
FROM JDEF OF ROUT.(1), (0) IS		J45		1W0=(0)	U131 000
A J, (1) HAS W-REGION INPUTS		J21		1W1=(1)	U131 010
		J90			U131 020
		J124			U131 030
SET UP TALLY IN W2		20W2		1W2=TALLY	U131 040
		11W0			U131 050
		11W1			U131 060
COPY JDEF OF (1) AND		10X22			U131 070
ASSIGN COPY AS JDEF OF (0).		J10			U131 080
		J74			U131 090
		J136			U131 100
		10X22			U131 110
		J11			U131 120
LOCATE ARGS IN 1W0 AND 1W1	90	11W0			U131 130
FOR HN, N=TALLY, IN 1W1,		11W1			U131 140
		11W2			U131 150
		U113			U131 160
IF NONE, EXIT.		7091			U131 170
SAVE ARG. OF 1W1 IN W3.		52H0			U131 180
		J74			U131 190
		20W3			U131 200
		11W2			U131 210
ADD 1 TO TALLY		J125			U131 220
		30H0			U131 230
SAVE LOC. OF ARG OF 1W0 IN W4		60W4			U131 240
FIND ARG. OF 1W0		52H0			U131 250
COMPILE (FIND ARG.)		U132			U131 260
		11W0			U131 270
INSERT (FIND ARG.) AT FRONT		10X22			U131 280
OF JDEF OF 1W0		J10			U131 290
		92			U131 300
REPLACE ARG OF 1W0 WITH		11W3			U131 310
ARG OF 1W1		21W4	90		U131 320
	91	11W2			U131 330
ERASE TALLY AND EXIT		J9	J35		U131 340
INSERT COMPILED (FIND ARG.)	92	44H0			U131 350
DIVIDE JDEF		J75			U131 360
SAVE REMAINDER		20W5			U131 370
		J6			U131 380
ADD (FIND ARG.) TO HEAD		J76			U131 390
ADD REMAINDER		11W5			U131 400
		J76	J8		U131 410
COMPILE (FIND ARG) (0). FINAL	U132	40H0			U132 000
ARGS. ARE DESIG. SYMBS. OUTPUT (0)		J90			U132 010
IS LIST OF INSTRS. WITHOUT TERM.		J136			U132 020
		40W1			U132 030
CREATE OUTPUT LIST IN W1.		20W1			U132 040
		10X32			U132 050
TEST IF ARGUMENT		J10			U132 060
IS DETERMINER.		10X37			U132 070
		J2			U132 080
ARGUMENT NOT A DETERMINER,		70	90		U132 090

SET UP Q-TALLY IN WO.		J90		U132 100
		J124		U132 110
		J50		U132 120
SAVE ARGUMENT	92	40H0		U132 130
FIND ITS DETERMINER		10X37		U132 135
		J10		U132 140
	94	J60		U132 145
		7093		U132 150
TALLY X151S=Q		12H0		U132 155
OF DETERMINER IN WO		10X151		U132 160
		J2		U132 170
		7094		U132 180
		11W0		U132 190
		J125		U132 200
COPY SYMBOL OF ARGUMENT	93	30H0	94	U132 210
		30H0		U132 220
		10X45		U132 225
		J10		U132 230
		J74		U132 240
MAKE INSTRUCTION 1QS.		11W0		U132 250
		10N1		U132 260
		U100		U132 270
INSERT INSTR. AT FRONT		11W1		U132 300
OF LIST 1W1.		J6		U132 310
EXIT, LEAVING 1W1.		J64		U132 320
		11W1	J31	U132 330
SET UP INSTR. SUBLIST,	90	J90		U132 340
AND MARK IT LOCAL,		J136		U132 350
AND SAVE IT IN WO.		J50		U132 360
SAVE DETERMINER		40H0		U132 363
FIND ITS DSCN		10X20		U132 365
		J10		U132 366
FIND LIST OF ARGUMENTS		10X31		U132 380
AND PROCESS THEM.		J10		U132 390
		10910		U132 400
		J100		U132 410
FIND JDEF		10X22		U132 420
		J10		U132 430
COPY IT		J74		U132 432
		J136		U132 433
		40H0		U132 434
FIND LAST INSTRUCTION		J61		U132 435
		52H0		U132 436
ERASE ITS LINK		10X46		U132 437
		J14		U132 438
		11W0		U132 440
FIND END OF LIST		J61		U132 445
INSERT LIST ON MAIN LIST		95		U132 460
INSERT SUBLIST IN LIST		11W0		U132 470
		11W1		U132 480
INSERT LIST ON MAIN LIST		95		U132 490
BRING IN LIST AND EXIT.		11W1	J31	U132 500
COMPILE (FIND ARG.)	910	U132		U132 510
AND INSERT IN SUBLIST.		11W0		U132 520

INSERT LIST ON MAIN LIST		95	J4	U132	530
INSERT LIST AT FRONT OF MAIN LIST	95	44H0		U132	540
		J75		U132	550
		40W2		U132	560
		20W2		U132	570
		J6		U132	580
		J76		U132	590
		11W2		U132	600
		J76		U132	610
		30W2	J8	U132	620
	U133	J45		U133	000
COMPILE JDEF WITH NAME (2)		10X199		1W4=NAME	U133 005
LIST OF SEGMENTS (1) AND FLOW		J73		1W3=SEG L	U133 010
CHART (0). OUTPUT (0) IS JDEF		J136		1W2=FLOW	U133 020
NAME NEW JDEF		J90		1W1=TALLY	U133 030
		J136		1W0=JDEF	U133 040
		J24		U133	050
ASSIGN NAME TO FIRST INSTRUCTION		11W3		U133	053
		J81		U133	054
		J81		U133	055
		11W4		U133	056
		10X41		U133	057
		J11		U133	058
		11W2		U133	060
		40W3		U133	070
ATTACH SEGMENTS TO FLOW CHART	91	J60		U133	080
		7090		U133	090
		12H0		U133	100
		11W3		U133	110
		J60		U133	120
		70J7		U133	130
		60W3		U133	140
		52H0		U133	145
		10X22		U133	150
		J11	91	U133	160
ASSIGN ADDRESSES TO SEGMENTS	90	30H0		U133	170
		30W3		U133	180
		11W2		U133	190
		1092		U133	200
		J100		U133	210
		11W0		U133	215
ASSEMBLE SEGMENTS IN NEW JDEF		11W3		U133	220
		10910		U133	230
		J100		U133	235
		40H0		U133	240
		U122	J35	U133	245
	910	11W0		U133	251
		J6		U133	260
		J76		U133	262
		20W0	J4	U133	264
ASSIGN ADDRESSES TO SEGMENT	92	64W5		1W5=SEGM	U133 270
FIND JDEF OF SEGMENT		10X22		U133	280
		J10		U133	290
		11W5		U133	310

END P OF SEGMENT		10X43		U133 320
		J10		U133 330
TEST IF P=7		10N7		U133 340
		J2		U133 350
IF NOT, GO TO 93		7093		U133 360
IF P=7		11W5		U133 370
		40H0		U133 375
		925		U133 376
FIND SYMB OF SEGM		10X45		U133 380
		J10		U133 390
		40H0		U133 400
TEST IF SYMB IS 0		10N0		U133 410
		J2		U133 420
IF NOT, GO TO 94		7094		U133 430
IF SO, GO TO 93		30H0	93	U133 440
	94	11W2		U133 450
		J6		U133 460
FIND SYMB ON FLOW		920		U133 470
		10X22		U133 480
FIND ITS JDEF		J10		U133 490
		70J7		U133 500
BRING IN (FIELD)=(SYMB)		10X45		U133 510
		11W1		U133 520
		J60		U133 530
		60W1		U133 540
		52H0		U133 545
LINK SEGM TO SYMB		U120		U133 550
		11W5		U133 560
		10X22		U133 570
BRING IN JDEF OF SEGM		J10		U133 580
FIND LINK OF SEGM	93	11W5		U133 590
		10X46		U133 600
		J10		U133 610
		40H0		U133 620
TEST IF LINK IS 0		10N0		U133 630
		J2		U133 640
		7095		U133 650
IF IT IS, EXIT		30H0	J8	U133 660
IF LINK IS NOT 0	95	11W2		U133 670
		J6		U133 680
FIND SYMB ON FLOW		920		U133 690
		10X22		U133 700
FIND ITS JDEF		J10		U133 710
		70J7		U133 720
BRING IN (FIELD)=(LINK)		10X46		U133 730
		11W1		U133 740
		J60		U133 750
		60W1		U133 755
		52H0		U133 756
LINK SEGM TO SYMB		U120	J4	U133 760
FIND SYMB ON FLOW	920	04J50		U133 765
		10921		U133 770
		J100	J30	U133 775
FIND NAME OF SYMB	921	40H0		U133 780

COMPARE WITH 1W0		10X41		U133	785
		J10		U133	790
		11W0		U133	795
		J2		U133	800
EXIT, STOP GENERATOR		70	J3	U133	805
EXIT, CONTINUE GENERATOR		30H0	J4	U133	810
	925	14X22		U133	815
		J10		U133	820
		J90		U133	825
		J136		U133	830
		40H0		U133	835
		10N7		U133	840
		10X43		U133	845
		J11	J65	U133	850
COMPILE (0) FROM ITS DSCN	U134	J50		U134	000
FIND LIST OF COMPILED ROUTINES		10X196		U134	010
AND SEEK SOURCE ROUTINE		1090		U134	020
		J100		U134	030
		70	J30	U134	040
IF FOUND, COMPILE (0)		11W0		U134	050
FROM DSCN OF XOURCE		U130	J30	U134	060
FIND SOURCE ROUTINE FOR (0)	90	40H0		U134	070
		10X20		U134	080
FIND PROCESS OF DSCN		J10		U134	090
		7091		U134	095
OF SOURCE		10X30		U134	100
		J10		U134	110
		7091		U134	115
		11W0		U134	120
		10X20		U134	130
		J10		U134	140
		10X30		U134	150
		J10		U134	160
COMPARE WITH PROCESS OF (0)		J2		U134	170
EXIT WITH SIGNAL		70	J3	U134	180
TO GENERATOR	91	30H0	J4	U134	190
COMPILE JDEF OF ROUTINE (0).	U135	J41		U135	000
OUT.(0) IS JDEF, IF EXISTS. SET H5.		60W0		U135	010
TEST IF IT EXISTS	90	10X22		U135	020
		J10		U135	030
IF SO, EXIT WITH H5+		70	J31	U135	040
		11W0		U135	050
FIND CLOSEST DEFCINATION		U136		U135	060
IF NONE, EXIT WITH H5-		70J31		U135	070
		40H0		U135	080
FIND AND APPLY		11W0		U135	090
RELEVANT PROCESS		U137		U135	100
		11W0		U135	110
FIND CLOSEST DEFINITION		U136		U135	120
TEST PROGRESS		U138		U135	140
EXIT OR REPEAT.		70J31		U135	150
		11W0	90	U135	160
FIND CLOSEST DEF. OF ROUTINE (0)	U136	J50		U136	000
		11W0		U136	010

		10X20		U136 020
		J10		U136 030
		70	90	U136 040
		11W0		U136 050
		10X24		U136 060
		J10		U136 070
		70J30	91	U136 080
	90	30H0		U136 090
		10X20	J30	U136 100
	91	30H0		U136 110
		10X24	J30	U136 120
FIND AND APPLY RELEVANT PROCESS	U137	J6		U137 000
TO ROUTINE (0) WITH DEF. (1)		10X198		U137 010
NO OUTPUT		J6		U137 020
		J10		U137 030
		J1	0	U137 040
TEST IF DEF (0) IS CLOSER TO	U138	10X197		U138 000
JDEF THAN DEF (1)		J6		U138 010
		J62		U138 020
		J6		U138 030
		J62	0	U138 040
COMPOSE FLOW CHART OF ROUTINE (0)	U139	J42		U139 000
FROM ITS JDEF		60W0		U139 010
		J90		U139 020
		J136		U139 030
STORE NAME OF CHART IN W1		60W1		U139 040
		10X26		U139 050
ASSIGN IT TO ROUTINE 1W0		J11		U139 060
		11W0		U139 070
		10X22		U139 080
FIND JDEF OF ROUTINE		J10		U139 090
COPY JDEF, MARK IT LOCAL		J74		U139 100
		J136		U139 110
STORE IT IN W2	91	60W2		U139 120
		J90		U139 130
CREATE SEGMENT		J136		U139 140
AND STORE IT		40H0		U139 150
		11W1		U139 160
INSERT SEGMENT		J6		U139 170
AT END OF FLOW CHART		J65		U139 180
		J6		U139 190
MAKE JDEF THE		10X22		U139 200
X22 OF SEGM.		J11		U139 210
		11W2		U139 220
FIND NEXT INSTRUCTION	92	J60		U139 230
		7094		U139 240
		12H0		U139 260
FIND ITS P		10X43		U139 270
		J10		U139 280
		7090		U139 285
TEST IF P=7		10N7		U139 290
		J2		U139 300
IF SO, DIVIDE SEGMENT		7090	93	U139 310
AND REPEAT.	93	J75	91	U139 320

IF NOT,	90	12H0		U139 330
		10X46		U139 340
		J10		U139 350
TEST IF LINK IS LOCAL		7092		U139 360
IF SO,		10X33		U139 370
DIVIDE SEGMENT AND REPEAT.		J10		U139 380
		10N9		U139 390
		J2		U139 400
IF NOT, FIND NEXT INSTRUCTION		7092	93	U139 410
	94	30H0		U139 420
		11W1		U139 430
LOCATE NEXT SEGMENT	98	J60		U139 440
IF NONE, EXIT		7099		U139 450
		12H0		U139 460
PUT ITS NAME IN W0		60W0		U139 470
		10X22		U139 480
FIND ITS JDEF		J10		U139 490
AND PUT IT IN W2		60W2		U139 500
FIND FIRST INSTRUCTION		J81		U139 510
		10X41		U139 520
FIND ITS NAME		J10		U139 530
IF NONE, SKIP TO 95		7095		U139 540
		11W0		U139 550
		J6		U139 560
ASSIGN IT AS SEGMENT NAME		10X41		U139 570
		J11		U139 580
	95	11W2		U139 590
FIND LAST INSTRUCTION		J61		U139 600
		52H0		U139 610
STORE IT IN W2		60W2		U139 620
		10X43		U139 630
TEST IF ITS P=7		J10		U139 640
		7096		U139 645
		10N7		U139 650
		J2		U139 660
IF NOT, SKIP TO 96		7096		U139 670
IF SO, SET P=7 IN SEGMENT		11W0		U139 680
		10N7		U139 690
		10X43		U139 700
		J11		U139 710
ASSIGN SYMB. OF INSTRUCTION TO SEGMENT		11W0		U139 720
		11W2		U139 730
		10X45		U139 740
		J10		U139 750
		10X45		U139 760
		J11	97	U139 770
SET P=0 IN SEGMENT	96	11W0		U139 780
		10N0		U139 790
		10X43		U139 800
		J11	97	U139 810
ASSIGN LINK OF INSTRUCTION TO SEGMENT	97	11W0		U139 820
		11W2		U139 830
		10X46		U139 840
		J10		U139 850

		10X46		U139	860
		J11	98	U139	870
EXIT	99	30H0	J32	U139	880
COMPILE JDEF OF ROUTINE (0)	U140	J50		U140	000
FROM ITS SDSC. NO OUTPUT		11W0		U140	001
COMPILE IT		930		U140	003
ASSIGN A NAME TO IT		910		U140	005
ASSIGN A TERMINATION TO IT		920	J30	U140	007
	930	J42		U140	009
FROM ITS SDSC. NO OUTPUT.		60W0	1W0=(0)	U140	010
		10X24		U140	020
FIND SDSC.		J10		U140	030
FIND INPUT-OUTPUT DIFFERENCE.		X7		U140	040
		40H0		U140	050
		10X50		U140	055
		J10		U140	056
TEST FOR NO DIFFERENCE.		10X80		U140	060
		J2		U140	070
		7090		U140	080
IF NO DIFFERENCE		J32		U140	090
EXIT WITH H5=-		J8	J3	U140	095
IF DIFFERENCE EXISTS,	90	11W0		U140	100
PRODUCE NEW ROUTINE WITHOUT DIFF.,		U141		U140	110
STORE IT IN W1.		60W1		U140	120
CONSTRUCT SDSC FOR A NEW ROUTINE,		11W0		U140	130
I(1W0) I(1W1).		U144		U140	140
SAVE NEW ROUTINE		60W2		U140	150
COMPILE ITS JDEF		930		U140	160
IF NULL, SKIP TO 91		7091		U140	165
BRING IN NEW ROUTINE		11W2		U140	170
PREFIX JDEF OF NEW ROUTINE TO JDEF		11W1		U140	180
OF 1W1, AND MODIFY SDSC OF 1W1.		U145		U140	190
	91	11W2		U140	195
ERASE PREFIXED ROUTINE		J72		U140	200
		11W0		U140	210
CONSTRUCT SDSC FOR A NEW ROUTINE,		11W1		U140	220
O(1W1) O(1W0).		U146		U140	230
SAVE NEW ROUTINE		60W2		U140	240
COMPILE ITS JDEF.		930		U140	250
IF NULL, SKIP TO 92		7092		U140	255
BRING IN NEW ROUTINE		11W2		U140	260
SUFFIX JDEF OF NEW ROUTINE TO JDEF		11W1		U140	270
OF 1W1, AND MODIFY SDSC OF 1W1.		U147		U140	280
ERASE SUFFIXED ROUTINE	92	11W2		U140	285
		J72		U140	290
BRING IN 1W0.		11W0		U140	300
FIND JDEF OF 1W1,		11W1		U140	310
		10X22		U140	320
		J10		U140	330
COPY IT.		J74		U140	340
		J136		U140	350
ASSIGN COPY AS JDEF OF 1W0.		10X22		U140	360
		J11		U140	370
		11W1		U140	380

ERASE 1W1		J72		U140 390
AND EXIT WITH H5=+		J32	J4	U140 400
	910	11W0		U140 410
FIND JDEF.		10X22		U140 420
		J10		U140 430
FIND FIRST INSTRUCTION.		J81		U140 440
		40H0		U140 450
		10X41		U140 455
		J10		U140 456
IF IT HAS NO NAME,		70	911	U140 460
		11W0		U140 470
ASSIGN IT V(X25) OF ROUTINE.		10X25		U140 480
		J10		U140 490
		10X41	J11	U140 500
IF IT HAS NAME, EXIT.	911	30H0	J8	U140 510
	920	11W0		U140 520
		10X22		U140 530
		J10		U140 540
		40H0		U140 550
TEST IF LAST INSTRUCTION		J61		U140 560
HAS A LINK.		52H0		U140 570
		10X46		U140 580
		J10		U140 590
IF NOT, ASSIGN LINK 0		70U122		U140 600
IF SO, EXIT.		30H0	J8	U140 610
PRODUCE ROUTINE FROM (0)	U141	J44		(0)=1W0 U141 000
WITHOUT DIFFERENCE (1).		J21		(1)=1W1 U141 010
OUTPUT (0) IS NEW ROUTINE.		11W1		U141 020
FIND ROUTINE RELEVANT TO DIFF.		U150		U141 030
		J74		U141 040
COPY IT AND MARK COPY LOCAL.		J136		U141 050
PUT RELEVANT ROUT. IN W2		60W2		U141 060
		10X24		U141 070
		J10		U141 080
PUT ITS SDSC IN W3		60W3		U141 090
SAVE IT		40H0		U141 100
LOCATE ITS VARIABLE CELL.		U152		U141 110
IF NONE, GO TO 90, AFTER J8		7093		U141 120
FIND ITS NAME		52H0		U141 130
FIND ITS VARIABLES		10X41		U141 135
		J10		U141 136
		11W1		U141 140
		10X41		U141 150
		J10		U141 160
FIND VARIABLE OF DIFFERENCE,		U153		U141 170
SUBSTITUTE IT FOR VAR. CELL NAME.	90	11W3		U141 180
BRING IN SDSC OF REL. ROUT.		11W1		U141 190
FIND VARIABLE OF DIFF.		10X41		U141 200
		J10		U141 210
LOCATE CELL OF SDSC WITH THIS NAME		U151		U141 220
IF NONE, GO TO 91.		7091		U141 230
		52H0		U141 232
		10X41		U141 234
		J10		U141 235

BRING IN SDSC OF REL. ROUT.		11W3		U141	240
		11W0		U141	250
FIND SDSC OF 1W0		10X24		U141	260
		J10		U141	270
SUB. NAMES OF (0) FOR VAR. OF W3		U154		U141	280
	91	11W0		U141	290
		10X24		U141	300
		J10		U141	310
FIND LIST OF AFFECTED CELLS OF 1W0		10X71		U141	320
		J10		U141	330
LOCATE NEXT CELL	92	J60		U141	340
		7094		U141	350
FIND CELL		12H0		U141	360
FIND ITS NAME		10X41		U141	370
		J10		U141	380
STORE IT IN W4		60W4		U141	390
FIND W3		11W3		U141	400
		J6		U141	410
LOCATE CELL OF W3 WITH SAME NAME		U151		U141	420
IF NONE, FIND NEXT.		7092		U141	430
		30H0		U141	440
		11W4		U141	450
		11W3		U141	460
SUBSTITUTE NAMES OF SDSC (W0) FOR		11W0		U141	470
VARIABLES OF (W3) IN CELL (W4)		10X24		U141	480
		J10		U141	490
		U154	92	U141	500
	95	11W1		U141	510
FIND ROUTINE RELEVANT TO W1,		U150		U141	520
FIND MODIFIED COPY.		11W2		U141	530
MAKE JDEF OF (0) FROM (1).		U155		U141	540
LEAVE NEW ROUTINE		11W2	J34	U141	550
	93	30H0		U141	560
		30H0	90	U141	565
	94	30H0	95	U141	570
CONSTRUCT SDSC FOR A NEW ROUTINE	U142	J53		U142	000
(0) OF (1) TO (2) OF (3). (0)=NAME		11W1		U142	010
		10X24		U142	015
		J10		U142	016
OF NEW ROUTINE.		10X71		U142	020
		J10		U142	030
COPY X71 OF 1W1		J74		U142	035
		J136		U142	036
REPLACE 1W1 WITH ITS X71		20W1		U142	040
		11W3		U142	050
		10X24		U142	055
		J10		U142	056
		10X71		U142	060
		J10		U142	070
		J74		U142	075
		J136		U142	076
REPLACE 1W3 WITH ITS X71		60W3		U142	080
		11W2		U142	090
ERASE HALF OF 1W2		U143		U142	100

		11W1		U142	110
		11W0		U142	120
ERASE HALF OF 1W0		U143		U142	130
		11W0		U142	140
		10X75		U142	150
TEST IF 1W1 NEEDS REVERSAL		J2		U142	160
		70	90	U142	170
		11W1		U142	180
IF SO, REVERSE		11W0		U142	190
		U148	90	U142	200
	90	11W2		U142	210
		10X76		U142	220
TEST IF 1W3 NEEDS REVERSAL		J2		U142	230
		70	91	U142	240
		11W3		U142	250
IF SO, REVERSE		11W2		U142	260
		U148	91	U142	270
COMBINE LISTS OF AFFECTED CELLS	91	11W3		U142	280
INTO NEW LIST.		11W1		U142	290
		U149		U142	300
ASSIGN COMBINED LIST		J90		U142	310
AS X71 OF NEW ROUTINE.		60W0		U142	320
		J6		U142	330
		J90		U142	332
		J136		U142	334
		60W2		U142	336
		J6		U142	338
		10X71		U142	340
		J11		U142	350
		11W2		U142	352
		10X24		U142	354
		J11		U142	356
		11W0	J33	U142	360
ERASE ATTRIBUTE NOT-(0) OF	U143	10X75		U143	000
ITEMS OF (1). NO OUTPUT		J2		U143	010
		7090		U143	020
FIND ATTRIBUTE NOT-(0).		10X76	91	U143	030
	90	10X75	91	U143	040
	91	J50		U143	050
ERASE IT.		1092		U143	060
		J100	J30	U143	070
	92	11W0		U143	080
		J14	J4	U143	090
CONSTRUCT SDSC FOR A NEW ROUTINE	U144	J50		U144	000
X75(0) TO X75(1)=OUTPUT ROUTINE		10X75		U144	010
		11W0		U144	020
		10X75		U144	030
		U142	J30	U144	040
PREFIX JDEF OF (1) TO JDEF OF (0)	U145	J51		U145	000
IN (0), AND MODIFY SDSC.		11W0		U145	010
		10X22		U145	020
FIND JDEF OF (1),		J10		U145	030
SAVE IT.		40H0		U145	040
DIVIDE IT AFTER HEAD,		J75		U145	050

SAVE TAIL IN W2.		40W2		U145 060
		20W2		U145 070
		11W1		U145 080
FIND JDEF OF (0),		10X22		U145 090
		J10		U145 100
		J74		U145 105
		J136		U145 106
INSERT IT AFTER HEAD OF JDEF (1).		J76		U145 110
		11W2		U145 120
ADD TAIL.		J76		U145 130
		30H0	J32	U145 135
CONSTRUCT SDSC FOR A NEW ROUTINE	U146	J50		U146 000
X76(0) TO X76(1)=OUTPUT ROUTINE		10X76		U146 010
		11W0		U146 020
		10X76		U146 030
		U142	J30	U146 040
SUFFIX JDEF OF (1) TO JDEF OF	U147	J51		U147 000
(0) IN (0), AND MODIFY SDSC		11W0		U147 010
		10X22		U147 020
FIND JDEF OF (0)		J10		U147 030
LOCATE ITS LAST SYMBOL		J61		U147 040
		11W1		U147 050
		10X22		U147 060
FIND JDEF OF (1)		J10		U147 070
		J74		U147 075
		J136		U147 076
		J76		U147 080
INSERT AFTER JDEF OF (0).		30H0	J31	U147 085
REVERSE ATTRIBUTES (0) OF ITEMS	U148	40H0		U148 000
ON (1).		10X75		U148 010
		J2		U148 020
		70	90	U148 030
FIND ATTRIBUTE OPPOSITE TO (0).		10X75	91	U148 040
	90	10X76	91	U148 050
STORE (0) IN W1, OPPOSITE IN W0.	91	J51		U148 060
GENERATE ITEMS AND REVERSE.		1092		U148 070
		J100	J31	U148 080
	92	40H0		U148 090
		40H0		U148 100
		11W1		U148 110
FIND VALUE OF ATTRIBUTE,		J10		U148 120
COPY IT,		J74		U148 130
AND ASSIGN AS VALUE OF OPPOSITE		J136		U148 140
		11W0		U148 145
		J11		U148 150
ERASE ATTRIBUTE.		11W1	J14	U148 160
COMBINE LISTS OF AFFECTED CELLS,	U149	J90		U149 000
(0)=LX75,(1)=LX76,OUTPUT=NEW LIST.		J136		U149 010
1W0=NEW LIST, 1W1=LX75,		J46		U149 020
1W2=LX76.		J22		U149 030
		11W1		U149 040
COMBINE PAIRS		1090		U149 050
		J100		U149 060
		11W1		U149 070

ADD REMAINDER OF LX75		1091		U149 080
		J100		U149 090
		11W2		U149 100
ADD REMAINDER OF LX76		1092		U149 110
		J100		U149 120
		11W0	J36	U149 130
COMBINE PAIRS	90	24W3		U149 140
		11W2		U149 150
GENERATE MEMBERS OF LX76		1093		U149 160
		J100	J4	U149 170
	93	24W4		U149 180
		11W3		U149 190
		10X41		U149 200
FIND SYMBOL OF LX75		J10		U149 210
		11W4		U149 220
		10X41		U149 230
FIND SYMBOL OF LX76		J10		U149 240
COMPARE FOR EQUALITY.		U107		U149 250
IF UNEQUAL TRY NEXT OF LX76.		70J4		U149 260
IF EQUAL,		11W1		U149 270
DELETE FROM LX75,		11W3		U149 280
		U169		U149 290
		11W2		U149 300
DELETE FROM LX76.		11W4		U149 310
		U169		U149 320
		11W4		U149 330
		40H0		U149 340
ADD LX75 TO LX76,		11W3		U149 350
		10X75		U149 360
		J10		U149 370
		10X75		U149 380
		J11		U149 390
IF PAIR IDENTICAL, DELETE		910		U149 395
		70	915	U149 396
AND INSERT PAIR IN NEW LIST.		11W0		U149 400
		J6		U149 410
STOP LX76 GENERATOR.		J64	J3	U149 420
ADD REMAINDER OF LX75	91	11W1		U149 430
		14X76		U149 435
		10X75		U149 440
		94	0	U149 450
ADD REMAINDER OF LX76	92	11W2		U149 460
		14X75		U149 465
		10X76		U149 470
		94	0	U149 480
ADD REMAINDER OF 1W3 TO 1W2	94	24W4		U149 490
		20W5		U149 500
		20W6		U149 505
		60W3		U149 510
		11W4		U149 520
		J10		U149 530
TEST FOR NORMAL FORM		96		U149 540
		7095		U149 550
IF NORMAL, DELETE	99	11W6		U149 560

		11W3		U149	570
		U169	0	U149	580
IF NOT NORMAL,	95	11W3		U149	590
CONSTRUCT PAIR ATTRIBUTE,		J90		U149	600
		J136		U149	610
		40H0		U149	620
		10X1		U149	630
INSERT IT,		J64		U149	640
		11W5		U149	650
		J11		U149	660
ADD TO 1W0		11W0		U149	670
AND DELETE FROM 1W3.		11W3		U149	680
		J64	98	U149	690
TEST IF NORMAL	96	44H0		U149	700
		J81		U149	710
		40H0		U149	720
TEST IF FIRST SYMBOL IS X1		10X1		U149	730
		J2		U149	740
IF SO, EXIT WITH H5+.		70	97	U149	750
TEST IF FIRST SYMBOL IS X2.		10X2		U149	760
		J2		U149	770
IF NOT, EXIT WITH H5-.		70J8		U149	780
IF SO, TEST IF SECOND IS X1		J82		U149	790
		10X1		U149	800
EXIT WITH H5 SET		J2	0	U149	810
	97	30H0	J8	U149	830
	98	J4	99	U149	840
	910	44H0		U149	845
		10X75		U149	850
		J10		U149	855
		40W0		U149	860
		20W0		U149	865
		40H0		U149	870
		10X76		U149	875
		J10		U149	880
	914	J60		U149	890
		70911		U149	895
		12H0		U149	900
		11W0		U149	905
		J60		U149	910
		70912		U149	915
		60W0		U149	920
		52W0		U149	925
		U107		U149	930
		70913	914	U149	935
	913	30H0	J30	U149	940
	912	30H0		U149	945
		30H0	913	U149	950
	911	30H0		U149	955
		11W0		U149	960
		J60		U149	965
		J5	913	U149	970
	915	30H0	J3	U149	975
FIND ROUTINE RELEVANT	U150	40H0		U150	000

TO DIFFERENCE (0)		10X50		U150	010
FIND TYPE OF DIFFERENCE		J10		U150	020
		10X90		U150	030
		J6		U150	040
FIND LIST OF RELEVANT ROUTINES		J10		U150	050
		J50		U150	060
		10X41		U150	065
		J10		U150	066
		10X32		U150	070
FIND TYPE OF AFFECTED CELL		J10		U150	080
		11W0		U150	090
		J6		U150	100
FIND RELEVANT ROUTINE		J10	J30	U150	110
LOCATE AFFECTED CELL (0) OF	U151	J50		U151	000
SDSC (1).		10X71		U151	010
FIND LIST OF AFFECTED CELLS		J10		U151	020
LOCATE NEXT CELL	90	J60		U151	030
IF NONE, EXIT WITH H5-		70J30		U151	040
		12H0		U151	050
FIND ITS NAME		10X41		U151	060
		J10		U151	070
		11W0		U151	080
COMPARE WITH (0).		U107		U151	090
EXIT OR CONTINUE		7090	J30	U151	100
LOCATE VARIABLE AFFECTED CELL	U152	10X71		U152	000
IN SDSC (0)		J10		U152	010
LOCATE NEXT CELL	90	J60		U152	020
IF NONE, EXIT WITH H5-		700		U152	030
		12H0		U152	040
FIND ITS NAME		10X41		U152	050
		J10		U152	060
FIND ITS TYPE		10X32		U152	070
		J10		U152	080
TEST IF VARIABLE		10X39		U152	090
		J2		U152	100
IF YES, EXIT WITH H5+		7090	J4	U152	110
SUBSTITUTE (0) FOR VARIABLE (1)	U153	J51		6W1=VAR.	U153 000
THROUGHOUT SDSC(2).		10X71		1W0=SUB.	U153 010
FIND LIST OF AFFECTED CLESS,		J10		U153	020
AND PROCESS IT.		1090		U153	030
		J100	J31	U153	040
SAVE CELL	90	40H0		U153	050
		10X41		U153	060
FIND ITS NAME		J10		U153	070
		11W1		U153	080
COMPARE IT WITH VARIABLE (1).		U107		U153	090
IF UNEQUAL, SKIP TO 91		7091		U153	100
IF EQUAL, SAVE CELL,		40H0		U153	110
ERASE ITS NAME,		10X41		U153	120
		J14		U153	130
AND SUBSTITUTE (0).		40H0		U153	140
		11W0		U153	150
		10X41		U153	160
		J11	91	U153	170

FIND INPUT OF CELL,	91	40H0		U153	180
		10X75		U153	190
		J10		U153	200
AND MAKE SUBSTITUTIONS.		92		U153	210
		10X76		U153	220
FIND OUTPUT OF CELL,		J10		U153	230
AND MAKE SUBSTITUTIONS.		92	J4	U153	240
SUBSTITUTE FOR CONTENTS OF LIST	92	4J60		U153	250
IF NO MORE CELLS, EXIT		70J8		U153	260
FIND FIRST SYMBOL		12H0		U153	270
AND COMPARE WITH 1W1		11W1		U153	300
		U107		U153	310
IF UNEQUAL, FIND NEXT		7092		U153	320
IF EQUAL		40W2		U153	330
		60W2		U153	340
MAKE SUBSTITUTION		11W0		U153	350
		J74		U153	355
		J136		U153	356
		21W2		U153	360
		30W2	92	U153	370
SUBSTITUTE NAMES OF SDSC (0) FOR	U154	J46		1W0=SDSC0	U154 000
VARIABLES OF SDSC (1) IN CELL		J22		1W1=SDSC1	U154 010
NAMED (2) OF (1).		11W0		1W2=CELL	U154 020
LOCATE CELL 1W2 IN SDSC 1W0		11W2		1W3=CELL0	U154 030
		U151		1W4=CELL1	U154 040
IF NONE, EXIT.		70J36		1W5=OUT.1	U154 050
PUT LOC. OF CELL OF W0 IN W3		20W3		U154	060
		11W1		U154	070
LOCATE CELL W2 IN SDSC W1		11W2		U154	080
		U151		U154	090
ERROR STOP		70J7		U154	100
PUT LOC OF CELL OF W1 IN W4		20W4		U154	110
		12W3		U154	120
		10X76		U154	130
FIND OUTPUT LIST OF W0 CELL		J10		U154	140
		12W4		U154	150
		10X76		U154	160
FIND OUTPUT LIST OF W1 CELL.,		J10		U154	170
AND STORE IN W5		20W5		U154	180
		90		U154	190
		12W3		U154	200
		10X75		U154	210
		J10		U154	215
		12W4		U154	220
		10X75		U154	230
		J10		U154	240
		20W5		U154	250
		90	J36	U154	260
LOCATE NEXT SYMBOL OF CELL 0	90	4J60		U154	270
IF NONE, EXIT		70J8		U154	280
		60W6		U154	285
		11W5		U154	290
LOCATE NEXT SYMBOL OF CELL 1		J60		U154	300
IF NONE, EXIT VIA 91		7091		U154	310

STORE LOCATION IN W5		60W5		U154 320
		12H0		U154 330
FIND TYPE OF SYMBOL OF 1		10X32		U154 340
		J10		U154 350
TEST IF A VARIABLE		10X39		U154 360
		J2		U154 370
IF NOT, GO TO 90 VIA 93		7092		U154 380
		52H0		U154 390
BRING IN SDSC 1		11W1		U154 420
		J6		U154 430
		12W6		U154 440
SUBSTITUTE THROUGH SDSC 1		U153	90	U154 470
	91	30H0	J8	U154 480
	92	30H0	90	U154 490
MAKE JDEF OF (0). (0) IS ROUTINE	U155	J47		U155 000
		J21		U155 005
WITH SDSC MODIFIED FROM (1).		11W0		U155 010
		11W1		U155 020
FIND JDEF OF (1)		10X22		U155 030
		J10		U155 040
		J74		U155 045
		J136		U155 046
ASSIGN IT AS JDEF OF (0)		10X22		U155 050
		J11		U155 060
BRING IN LISTS OF AFFECTED CELLS		93	91	U155 065
	93	04J0		U155 066
		11W0		U155 070
		10X24		U155 080
FIND SDSC OF (0)		J10		U155 090
		10X71		U155 100
FIND LIST OF AFFECTED CELLS		J10		U155 110
AND PUT IN W2		20W2		U155 120
		11W1		U155 130
		10X24		U155 140
FIND SDSC OF (1)		J10		U155 150
		10X71		U155 160
FIND LIST OF AFFECTED CELLS		J10		U155 170
AND PUT IN W3		60W3		U155 180
		11W2	0	U155 190
FIND NEXT CELL OF (0)	91	J60		U155 200
		7090		U155 210
		11W3		U155 220
FIND NEXT CELL OF (1)		J60		U155 230
		60W3		U155 240
		70J7		U155 250
		52H0		U155 260
		40H0		U155 220
FIND ITS NAME		10X41		U155 280
		J10		U155 290
TEST IF IT IS A VARIABLE		10X32		U155 300
		J10		U155 310
		10X39		U155 320
		J2		U155 330
		7092		U155 340

SO, STORE IN W4
STORE NAME OF SYMBOL OF (0) IN W5

SUBSTITUTE W5 FOR W4

THROUGHOUT (0)

SUBSTITUTE FOR VARIABLES IN CELLS

PUT X75 OF W2 IN W6

PUT X75 OF W3 IN W7
SUBS. FOR VARIABLE IN X75

PUT X76 OF W2 IN W6

PUT X76 OF W3 IN W7
SUBS. FOR VARIABLE IN X76

EXIT

FIND VARIABLES AND SUBSTITUTE

FIND NEXT IN W6
IF NONE, EXIT

	20W4		U155 350
	12H0		U155 360
	20W5		U155 370
	99	91	U155 375
99	04J0		U155 380
	11W0		U155 385
	10X22		U155 385
	J10		U155 386
	11W4		U155 390
	10X41		U155 395
	J10		U155 396
	11W5		U155 400
	10X41		U155 405
	J10		U155 406
	U156	0	U155 410
92	30H0	91	U155 420
90	30H0		U155 430
	30H0		U155 440
	93		U155 450
94	J60		U155 460
	7095		U155 470
	60W2		U155 480
	52H0		U155 490
	10X75		U155 500
	J10		U155 510
	20W6		U155 520
	J60		U155 530
	70J7		U155 540
	60W3		U155 550
	52H0		U155 560
	10X75		U155 570
	J10		U155 580
	20W7		U155 590
	96		U155 600
	12W2		U155 610
	10X76		U155 620
	J10		U155 630
	20W6		U155 640
	12W3		U155 650
	10X76		U155 660
	J10		U155 670
	20W7		U155 680
	96		U155 690
	11W3		U155 700
	11W2	94	U155 710
95	30H0		U155 720
	30H0	J37	U155 730
96	04J0		U155 740
	11W6		U155 745
	J60		U155 750
	70J8		U155 760
	60W6		U155 770
	11W7		U155 780
	J60		U155 790

TEST IF NEXT IN W7 IS VARIABLE

IF NOT, FIND NEXT
IF SO,
SUBSTITUTE W5 FOR W4
THROUGHOUT W0

SUBSTITUTE SYMBOL (0) FOR
VARIABLE (1) IN JDEF OF (2)
FIND NEXT LINE OF JDEF
IF NONE, EXIT

FIND ITS SYMB.

COMPARE SYMB WITH 1W1

IF UNEQUAL, FIND NEXT
IF EQUAL,
COPY 1W0 AS NEW SYMB
FOR LINE OF JDEF

DELETE CAREFULLY SYMBOL (0)
FROM LIST (1)

LOCATE (0) ON (1)

DIVIDE LIST AFTER LOC(0)
PUT REMAINDER IN W1
DELETE LAST SYMBOL FROM (1)

LOCATE LAST SYMBOL OF (1)

REATTACH REMAINDER

FIND BIGGEST DIFFERENCE
BETWEEN DSNS (0) AND (1)

	70J7		U155	800
	60W7		U155	810
	12H0		U155	820
	10X32		U155	830
	J10		U155	840
	10X39		U155	850
	J2		U155	860
	7098		U155	870
	52H0		U155	880
	20W4		U155	890
	52H0		U155	900
	20W5		U155	910
	11W0		U155	920
	10X22		U155	921
	J10		U155	922
	11W4		U155	924
	11W5		U155	926
	U156	96	U155	928
98	30H0		U155	930
	30H0	96	U155	940
U156	J52		U156	000
	11W2		U156	010
90	J60		U156	020
	7091		U156	030
	12H0		U156	040
	10X45		U156	050
	J10		U156	060
	11W1		U156	070
	U107		U156	080
	7090		U156	090
	12H0		U156	100
	11W0		U156	110
	J74		U156	120
	J136		U156	130
	10X45		U156	140
	J11	90	U156	210
91	30H0	J32	U156	220
U169	J41		U169	000
	20W0		U169	010
	40H0		U169	020
	11W0		U169	030
	J62		U169	040
	70J30		U169	050
	J75		U169	060
	20W1		U169	070
	40H0		U169	080
	J70		U169	090
	J61		U169	100
	11W1		U169	110
	J76		U169	120
	30H0	J31	U169	130
X5	J43		X5	000
	20W0		X5	010
	60W1		X5	020

		10X30		X5	030
FIND PROCESS OF (1)		J10		X5	040
		11W0		X5	050
		10X30		X5	060
FIND PROCESS OF (0)		J10		X5	070
COMPARE PROCESSES		J2		X5	080
IF EQUAL, COMPARE ARGUMENTS		70	90	X5	090
IF UNEQUAL, SIGNAL X62		10X62	J33	X5	100
FIND ARG. LISTS OF (0) AND (1)	90	910		X5	110
FIND NEXT ARGS. OF (0) AND (1)	92	911		X5	120
NONE, GO TO 91		7091		X5	130
TEST ARGUMENT TYPES		X6		X5	140
IF UNEQUAL, EXIT WITH SIGNAL		70J33	92	X5	150
ALL TYPES EQUAL, TEST ARGS.	91	910		X5	160
FIND NEXT ARGS, OF (0) AND (1)	93	911		X5	170
NONE, GO TO 94		7094		X5	180
		40H0		X5	190
		10X32		X5	200
FIND ARGUMENT TYPE		J10		X5	210
		70J7		X5	220
		10X37		X5	230
TEST IF DETERMINER		J2		X5	240
IF NOT, FIND NEXT ARG.		7093		X5	250
IF SO, FIND BIGGEST DIFFERENCE		X5		X5	260
SAVE DIFFERENCE SIGNAL		40H0		X5	270
TEST IF NO DIFFERENCE		10X55		X5	280
		J2		X5	290
IF DIFFERENCE, EXIT WITH SIGNAL		7097		X5	300
NO DIFFERENCE, FIND NEXT ARG.		30H0	93	X5	310
NO DETERMINER DIFFERENCES	94	910		X5	320
FIND NEXT ARGS. OF (0) AND (1)		911		X5	330
NONE, EXIT WITH X55		7095		X5	340
		40H0		X5	350
TEST IF NAMES		10X36		X5	360
		J2		X5	370
		7096		X5	380
SIGNAL DIFFERENT NAMES		10X66	J33	X5	390
SIGNAL DIFFERENT LOCATIONS	96	10X67	J33	X5	400
EXIT, NO DIFFERENCE	95	10X55	J33	X5	410
FIND ARGUMENT LISTS	910	11W0		X5	420
		10X31		X5	430
FIND LIST OF (0)		J10		X5	440
AND PUT IN W2.		20W2		X5	450
		11W1		X5	460
		10X31		X5	470
FIND LIST OF (1),		J10		X5	480
AND PUT IN W3.		20W3	0	X5	490
FIND NEXT ARGS. OF (0) AND (1)	911	11W2		X5	500
FIND NEXT OF (0)		J60		X5	510
IF NONE, SET H5 -		700		X5	520
		60W2		X5	530
		52H0		X5	540
		11W3		X5	550
FIND NEXT OF (1)		J60		X5	560

		70J7		X5	570
		60W3		X5	580
		52H0	0	X5	590
TEST TYPES OF ARGUMENTS (0), (1)	X6	J43		X6	000
OUTPUT (0) IS DIFFERENCE, SET H5.		20W0		X6	010
		60W1		X6	020
		10X32		X6	030
FIND ARGUMENT TYPE OF (1)		J10		X6	040
STORE IN W3		60W3		X6	045
		11W0		X6	050
		10X32		X6	060
FIND ARGUMENT TYPE OF (0)		J10		X6	070
STORE IN W2		60W2		X6	075
COMPARE ARGUMENT TYPES		J2		X6	080
		70	J33	X6	090
UNEQUAL, TEST IF (0) DETERMINER		11W2		X6	100
		10X37		X6	110
		J2		X6	120
		7091		X6	130
SIGNAL (0) IS DETERMINER		10X65	90	X6	140
TEST IF (0) IS A NAME	91	11W2		X6	150
		10X36		X6	160
		J2		X6	170
		7092		X6	180
SIGNAL (0) A NAME		10X68	90	X6	190
IF (0) A LOCATION, SIGNAL X69	92	10X69	J33	X6	200
	90	J5	J33	X6	210
FIND LARGEST DIFFERENCE	X7	J44		X7	000
BETWEEN INPUT AND OUTPUT		10X71		X7	010
OF SDSC (0). PUT DIFF. IN (0).		J10		X7	020
PUT LIST OF AFFECTED CELLS IN W0.		60W0		X7	030
FIND DIFFERENCE, NOT IN H0.		1090		X7	040
		J100		X7	050
EXIT IF FOUND		70J34		X7	060
		11W0		X7	070
FIND DIFFERENCE IN H0		1091		X7	080
		J100		X7	090
EXIT IF FOUND		70J34		X7	100
		J90		X7	110
IF NOT FOUND,		J136		X7	120
		40H0		X7	125
LEAVE NULL DIFFERENCE,		10X80		X7	130
V(X50)=X80, IN H0		10X50		X7	140
		J11	J34	X7	150
PUT AFFECTED CELL IN W1	90	64W1		X7	160
		10X41		X7	170
FIND TYPE OF AFFECTED CELL		J10		X7	180
COPY AND SAVE ITS NAME		J74		X7	182
		J136		X7	183
		40H0		X7	184
		10X32		X7	190
		J10		X7	200
TEST IF H0		10X35		X7	220
		J2		X7	230

IF SO, EXIT.		7092	J8	X7	240
IF NOT, CREATE DIFFERENCE,	92	J90		X7	250
		J136		X7	260
SAVE ITS NAME IN W2,		60W2		X7	270
		J6		X7	280
ASSIGN TYPE OF CELL		10X41		X7	290
		J11		X7	300
		11W1		X7	310
FIND TYPE OF DIFFERENCE		93		X7	320
		11W2		X7	330
		J6		X7	340
		10X50		X7	350
ASSIGN TYPE TO 1W2		J11		X7	360
BRING DIFFERENCE IN, AND QUIT		11W2	J3	X7	370
FIND TYPE OF AFFECTED CELL	91	64W1		X7	380
		10X41		X7	390
		J10		X7	400
COPY ITS NAME		J74		X7	410
		J136		X7	420
		J3	92	X7	430
SAVE AFFECTED CELL	93	44H0		X7	440
		10X75		X7	450
		J10		X7	460
FIND ITS FIRST INPUT SYMBOL,		J81		X7	470
AND SAVE IN W3.		20W3		X7	480
SAVE AFFECTED CELL		40H0		X7	490
		10X76		X7	500
		J10		X7	510
FIND ITS FIRST OUTPUT SYMBOL		J81		X7	520
		11W3		X7	530
COMPARE SYMBOLS		U107		X7	540
		7094		X7	550
IF IDENTICAL,		30H0		X7	560
CALL DIFFERENCE (COPY)		10X84	0	X7	570
NOT IDENTICAL,	94	40H0		X7	580
COUNT LIST X75		10X75		X7	590
		J10		X7	600
		J126		X7	610
		20W3		X7	620
COUNT LIST X76		10X76		X7	630
		J10		X7	640
		J126		X7	650
		60W4		X7	660
		11W3		X7	670
		J114		X7	680
IF LISTS EQUAL		7095		X7	690
CALL DIFFERENCE (REPLACE)		10X83	0	X7	700
	95	11W3		X7	710
		11W4		X7	720
		J115		X7	730
IF 1W4 LARGER,		7096		X7	740
CALL DIFFERENCE (ADD)		10X82	0	X7	750
IF SMALLER, (DELETE)	96	10X81	0	X7	760
DATA	5	1			

A
B
C
D
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H
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J
K
L
M
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(0)
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(2)
(3)
(4)

A0	0	0
L1	21A	
L2	21B	
L3	21C	
L4	21D	
L5	21E	
L6	21F	
L7	21G	
L8	21H	
L9	21I	
L10	21J	
L11	21K	
L12	21L	
L13	21M	
L14	21N	
L15	21O	
L16	21P	
L17	21Q	
L18	21R	
L19	21S	
L20	21T	
L21	21U	
L22	21V	
L23	21W	
L24	21X	
L25	21Y	
L26	21Z	
L30	210	
L31	211	
L32	212	
L33	213	
L34	214	
L35	215	
L36	216	
L37	217	
L38	218	
L39	219	
L40	21(0)	
L41	21(1)	
L42	21(2)	
L43	21(3)	
L44	21(4)	
N0	1	0000
N1	1	0001
N2	1	0002
N3	1	0003
N4	1	0004
N5	1	0005
N6	1	0006
N7	1	0007
N8	1	0008
N9	1	0009
N10	1	0010
N13	1	0013

A0	000
L1	000
L2	000
L3	000
L4	000
L5	000
L6	000
L7	000
L8	000
L9	000
L10	000
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L38	000
L39	000
L40	000
L41	000
L42	000
L43	000
L44	000
N0	000
N1	000
N2	000
N3	000
N4	000
N5	000
N6	000
N7	000
N8	000
N9	000
N10	000
N13	000

	N28	1	0028	N28	000
	N29	1	0029	N29	000
	N30	1	0030	N30	000
	N35	1	0035	N35	000
	N36	1	0036	N36	000
	N37	1	0037	N37	000
	N38	1	0038	N38	000
	N44	1	0044	N43	000
	N49	1	0049	N49	000
	N50	1	0050	N50	000
	N60	1	0060	N60	000
	N65	1	0065	N65	000
	N77	1	0077	N77	000
	N116	1	0116	N116	000
DIFFERENCE	X0	0	0	X0	000
REMAINDER OF PUSHDOWN LIST	X1	90	0	X1	000
	90	0		X1	010
		X32		X1	020
		X1		X1	030
		X33		X1	040
		L18		X1	050
		X34		X1	060
		N0	0	X1	070
VARIABLE S1	X2	90	0	X2	000
	90	0		X2	010
		X32		X2	020
		X38		X2	030
		X33		X2	040
		L19		X2	050
		X34		X2	060
		N1	0	X2	070
VARIABLE S0	X3	90	0	X3	000
	90	0		X3	010
		X32		X3	020
		X39		X3	030
		X33		X3	040
		L19		X3	050
		X34		X3	060
		N0	0	X3	070
VARIABLE S2	X4	90	0	X4	000
	90	0		X4	010
		X32		X4	020
		X39		X4	030
		X33		X4	040
		L19		X4	050
		X34		X4	060
		N2	0	X4	070
DSCN (DESCRIPTIVE NAME)	X20	0	0	X20	000
LDEF (L-DEFINITION)	X21	0	0	X21	000
JDEF (J-DEFINITION)	X22	0	0	X22	000
ASOJ (ASSOCIATED J)	X23	0	0	X23	000
SDSC OF ROUTINE	X24	0	0	X24	000
IPLN OF ROUTINE	X25	0	0	X25	000
PROCESS	X30	0	0	X30	000

LIST OF ARGUMENTS	X31	0	0	X31	000
TYPE (OF ARGUMENT)	X32	0	0	X32	000
REGION	X33	0	0	X33	000
LOCATION	X34	0	0	X34	000
MEMBER OF HO LIST	X35	0	0	X35	000
NAMED SYMBOL	X36	0	0	X36	000
DETERMINER	X37	0	0	X37	000
UNKNOWN CONSTANT	X38	0	0	X38	000
VARIABLE	X39	0	0	X39	000
TYPE (OF IPL WORD)	X40	0	0	X40	000
NAME	X41	0	0	X41	000
SIGN	X42	0	0	X42	000
P-PREFIX	X43	0	0	X43	000
Q-PREFIX	X44	0	0	X44	000
SYMB	X45	0	0	X45	000
LINK	X46	0	0	X46	000
TYPE OF DIFFERENCE	X50	0	0	X50	000
TEST FOR DIFFERENCE	X51	0	0	X51	000
NO DIFFERENCE	X55	0	0	X55	000
JDEF ABSENT	X60	0	0	X60	000
LDEF ABSENT	X61	0	0	X61	000
DSCN-WRONG FUNCTION NAME	X62	0	0	X62	000
DSCN-WRONG ARGUMENT TYPE	X63	0	0	X63	000
LDEF HAS CONNECTIVE	X64	0	0	X64	000
DSCN-DETERM. FOR OTHER	X65	0	0	X65	000
DSCN-ONE NAME FOR OTHER	X66	0	0	X66	000
DSCN-ONE LOC FOR OTHER	X67	0	0	X67	000
DSCN-NAME FOR LOCATION	X68	0	0	X68	000
DSCN-LOCATION FOR NAME	X69	0	0	X69	000
OPERATOR ROUTINE	X70	0	0	X70	000
LIST OF AFFECTED CELLS OF SDSC	X71	0	0	X71	000
INPUT STATE OF CELL	X75	0	0	X75	000
OUTPUT STATE OF CELL	X76	0	0	X76	000
SDSC - NO DIFFERENCE	X80	0	0	X80	000
SDSC - DELETION	X81	0	0	X81	000
SDSC - ADDITION	X82	0	0	X82	000
SDSC - REPLACEMENT	X83	0	0	X83	000
SDSC - COPY	X84	0	0	X84	000
INDEX TO RELEVANT ROUTINES	X90	90	0	X90	000
	90	0		X90	050
LIST OF ADD ROUTINES		X82		X90	100
		91		X90	110
LIST OF REPLACE ROUTINES		X83		X90	200
		92	0	X90	210
	91	93	0	X90	300
ADD.	93	0		X90	350
HO IS AFFECTED CELL		X35		X90	400
		X107	0	X90	410
	92	94	0	X90	500
REPLACE.	94	0		X90	550
NAMED CELL IS AFFECTED CELL		X36		X90	600
		X106	0	X90	610
ROUT. INSERT (1) AT END OF	X100	90	0	X100	000
VALUE LIST OF ATTRIBUTE	90	0		X100	010

(0) OF (2).		X25		X100 015
DSCN IS 91.		97		X100 016
		X20		X100 020
		91	0	X100 030
	91	92	0	X100 040
	92	0		X100 050
PROCESS IS X110		X30		X100 060
		X110		X100 070
		X31		X100 080
		93	0	X100 090
ARGS. ARE	93	0		X100 100
X120		X120		X100 110
X121		X121	0	X100 120
	97	99	0	X100 210
	99	0		X100 220
SYMB. = J13		X33		X100 230
		L10		X100 240
		X34		X100 250
		N13	0	X100 260
ROUT. INSERT (0) AT END OF (1)	X101	90	0	X101 000
	90	0		X101 010
		X20		X101 020
		91		X101 030
JDEF IS 94		X22		X101 035
		94	0	X101 036
	91	92	0	X101 040
	92	0		X101 050
PROCESS IS X110		X30		X101 060
		X110		X101 070
		X31		X101 080
		93	0	X101 090
	93	0		X101 100
		X122		X101 110
		X120	0	X101 120
	94	0		X101 130
		95	0	X101 140
	95	96	0	X101 150
	96	0		X101 160
J65		X45		X101 170
		97		X101 180
		X46		X101 190
		98	0	X101 200
	97	99	0	X101 210
PROCESS J65	99	0		X101 220
		X33		X101 230
		L10		X101 240
		X34		X101 250
		N65	0	X101 260
	98	910	0	X101 270
LINK 0	910	0		X101 280
		X33		X101 290
		N0	0	X101 300
ROUT. FIND V((0),(1))	X102	90	0	X102 000
	90	0		X102 010

TYPE--DETERMINER		X32		X102 020
		X37		X102 030
		X20		X102 040
JDEF IS 94		91		X102 050
		X22		X102 060
		94	0	X102 070
	91	92		X102 080
VERB		920	0	X102 085
	92	0		X102 090
		X30		X102 100
PROCESS IS X111. FIND VALUE		X111		X102 110
		X31		X102 120
		93	0	X102 130
ARGS. ARE	93	0		X102 140
X122. (0)		X122		X102 150
X120. (1)		X120	0	X102 160
	94	0		X102 170
		95	0	X102 180
	95	96	0	X102 190
	96	0		X102 200
		X45		X102 210
		97		X102 220
		X46		X102 230
		98	0	X102 240
	97	99	0	X102 250
PROCESS J10	99	0		X102 260
		X33		X102 270
		L10		X102 280
		X34		X102 290
		N10	0	X102 300
	98	910	0	X102 310
LINK 0	910	0		X102 320
		X33		X102 330
		N0	0	X102 340
VERB (FIND)	920	21FIND		X102 350
ROUT. TESTN IF (0) LESS THAN NO	X103	90	0	X103 000
	90	0		X103 010
		X20		X103 020
		91		X103 030
		X22		X103 040
		94	0	X103 050
	91	92	0	X103 060
	92	0		X103 070
		X30		X103 080
PROCESS IS X112 TESTN, LESS		X112		X103 090
		X31		X103 100
		93	0	X103 110
ARGUMENTS	93	0		X103 120
(0)		X122		X103 130
NO		X124	0	X103 140
	94	0	0	X103 150
ROUT. TESTN IF (0) LESS THAN (1)	X104	90	0	X104 000
	90	0		X104 010
		X20		X104 020

	91		X104 030
	X22		X104 040
	94	0	X104 050
91	92	0	X104 060
92	0		X104 070
	X30		X104 080
	X112		X104 090
	X31		X104 100
	93	0	X104 110
93	0		X104 120
	X122		X104 130
	X120	0	X104 140
94	0		X104 150
	95	0	X104 160
95	96	0	X104 170
96	0		X104 180
	X45		X104 190
	97		X104 200
	X46		X104 210
	98	0	X104 220
97	99	0	X104 230
99	0		X104 240
	X33		X104 250
	L10		X104 260
	X34		X104 270
	N116	0	X104 280
98	910	0	X104 290
910	0		X104 300
	X33		X104 310
	N0	0	X104 320
J3. REPLACE TOP OF H5 WITH J3.	X105	90	0
	90	0	X105 000
	X20		X105 010
	920		X105 015
	X25		X105 016
	91		X105 020
	X24		X105 030
	92	0	X105 040
IPLN OF J3	91	93	0
	93	0	X105 050
	X33		X105 060
	L10		X105 070
	X34		X105 080
	N3	0	X105 090
SDSC OF J3	92	94	0
	94	0	X105 100
	X71		X105 110
	95	0	X105 120
	95	0	X105 130
	97	0	X105 140
AFFECTED CELL IS 97	97	98	0
	98	0	X105 150
	X41		X105 160
	X175		X105 170
			X105 180
			X105 190
			X105 200
			X105 210

		X75		X105 220
		99		X105 230
		X76		X105 240
		910	0	X105 250
INPUT STATE OF H5	99	0		X105 260
		X2		X105 270
		X1	0	X105 280
OUTPUT STATE OF H5.	910	0		X105 290
		911		X105 300
		X1	0	X105 310
	911	912	0	X105 320
	912	0		X105 330
		X32		X105 335
		X36		X105 336
		X33		X105 340
		L10		X105 350
		X34		X105 360
		N3	0	X105 370
	920	921	0	X105 380
DSCN	921	0		X105 390
		X30		X105 400
PROCESS IS X113		X113	0	X105 410
P2(C). REPLACE TOP OF C WITH	X106	90	0	X106 000
TOP OF H0.	90	0		X106 010
DSCN		X20		X106 012
		930		X106 013
		X22		X106 015
		920		X106 016
		X25		X106 020
		91		X106 030
		X24		X106 040
		92	0	X106 050
IPLN OF P2(C)	91	93	0	X106 060
	93	0		X106 070
		X33		X106 080
		L16		X106 090
		X34		X106 100
		N2	0	X106 110
SDSC OF P2(C)	92	94	0	X106 120
	94	0		X106 130
		X71		X106 140
		95	0	X106 150
	95	0		X106 160
AFFECTED CELLS ARE		96		X106 170
96 AND 97		97	0	X106 180
	96	98	0	X106 190
	98	0		X106 200
		X41		X106 210
		X170		X106 220
		X75		X106 240
		99		X106 250
		X76		X106 260
		910	0	X106 270
INPUT STATE OF H0	99	0		X106 280

		X3		X106 290
		X1	0	X106 300
OUTPUT STATE OF H0	910	0		X106 310
		X1	0	X106 320
	97	914	0	X106 330
	914	0		X106 340
		X41		X106 341
NAME IS VARIABLE, X4		X4		X106 342
		X75		X106 350
		915		X106 360
		X76		X106 370
		916	0	X106 380
INPUT STATE OF C	915	0		X106 390
		X2		X106 400
		X1	0	X106 410
OUTPUT STATE OF C	916	0		X106 420
		X3		X106 430
		X1	0	X106 440
	920	0		X106 450
		921	0	X106 460
	921	922	0	X106 470
	922	0		X106 480
		X43		X106 490
		N2		X106 500
		X45		X106 510
		X4	0	X106 520
	930	0		X106 530
		X30		X106 540
PROCESS IS X114.		X114		X106 550
		X31		X106 560
		931	0	X106 570
	931	0		X106 580
ARG. IS X125.		X125	0	X106 590
P1(S). ADD S TO TOP OF H0.	X107	90	0	X107 000
	90	0		X107 010
		X20		X107 012
		930		X107 013
		X22		X107 015
		920		X107 016
		X25		X107 020
		91		X107 030
		X24		X107 040
		92	0	X107 050
IPLN OF P1(S)	91	93	0	X107 060
	93	0		X107 070
		X33		X107 080
		L16		X107 090
		X34		X107 100
		N1	0	X107 110
SDSC OF P1(S)	92	94	0	X107 120
	94	0		X107 130
		X71		X107 140
		95	0	X107 150
	95	0		X107 160

		97	0	X107	170
AFFECTED CELL IS H0	97	98	0	X107	180
	98	0		X107	190
		X41		X107	200
		X170		X107	210
		X75		X107	220
		99		X107	230
		X76		X107	240
		910	0	X107	250
INPUT STATE OF H0	99	0		X107	260
		X1	0	X107	270
OUTPUT STATE OF H0	910	0		X107	280
		X3		X107	290
		X1	0	X107	300
	920	0		X107	310
		921	0	X107	320
	921	922	0	X107	330
	922	0		X107	340
		X43		X107	350
		N1		X107	360
		X45		X107	370
		X3	0	X107	380
DSCN	930	0		X107	390
		X30		X107	400
PROCESS IS X115.		X115		X107	410
		X31		X107	420
		931	0	X107	430
	931	0		X107	440
		X126	0	X107	450
ARG. IS X126	X110	0		X110	000
		911		X110	010
		912		X110	020
		X31		X110	030
		914		X110	040
		915		X110	050
		X31	0	X110	060
	911	21INSER		X110	001
	912	21T		X110	002
	914	21AT EN		X110	004
	915	21D OF		X110	005
	X111	0		X111	000
		911		X111	010
		912		X111	020
		913		X111	030
		914		X111	040
		915		X111	050
		916		X111	060
		X31		X111	070
		918		X111	080
		X31	0	X111	090
	911	21THE V		X111	001
	912	21ALUE		X111	002
	913	21-OF T		X111	003
	914	21HE AT		X111	004

	915	21TRIBU		X111	005
	916	21TE		X111	006
	918	21OF		X111	008
PROCESS TESTN () LESS ()	X112	0		X112	000
		911		X112	010
		912		X112	020
		X31		X112	030
		914		X112	040
		915		X112	050
		916		X112	060
		X31	0	X112	070
	911	21TEST-		X112	080
	912	21N IF		X112	090
	914	21IS LE		X112	100
	915	21SS TH		X112	110
	916	21AN		X112	120
PROCESS. SET H5 -	X113	0		X113	000
		911		X113	010
		912		X113	020
		913	0	X113	030
	911	21SET H		X113	040
	912	215 MIN		X113	050
	913	21US		X113	060
PROCESS. OUTPUT (0) TO ().	X114	0		X114	000
		911		X114	010
		912		X114	020
		913		X114	030
		X31	0	X114	040
	911	21OUTPU		X114	050
	912	21T (0)		X114	060
	913	21 TO		X114	070
PROCESS. INPUT ().	X115	0		X115	000
		911		X115	010
		X31	0	X115	020
	911	21INPUT		X115	030
ARG. X120 (1)	X120	90		X120	000
		92	0	X120	005
	90	0		X120	010
		X32		X120	020
TYPE- HQ LIST LOCATION		X35		X120	030
		X37		X120	040
		91		X120	050
		X45		X120	060
		X170	0	X120	070
	91	0		X120	080
		X151		X120	090
		X150	0	X120	100
	92	21(1)		X120	110
ARG. X121	X121	90	0	X121	000
	90	0		X121	010
		X32		X121	020
TYPE- DETERMINER		X37		X121	030
		X22		X121	035
JDEF IS 94		94		X121	036

DSCN IS 91.

PROCESS IS X111.

ARGS. ARE
X122
X123

ARG. X122 (0)

TYPE- H0 LIST LOCATION

ARG. X123 (2)

TYPE- H0 LIST LOCATION

	X20		X121 040
	91	0	X121 050
91	92	0	X121 060
92	0		X121 070
	X30		X121 080
	X111		X121 090
	X31		X121 100
	93	0	X121 110
93	0		X121 120
	X122		X121 130
	X123	0	X121 140
94	0		X121 150
	95	0	X121 160
95	96	0	X121 170
96	0		X121 180
	X45		X121 190
	97		X121 200
	X46		X121 210
	98	0	X121 220
97	99	0	X121 230
99	0		X121 240
	X33		X121 250
	L10		X121 260
	X34		X121 270
	N10	0	X121 280
98	910	0	X121 290
910	0		X121 300
	X33		X121 310
	N0	0	X121 320
X122	90		X122 000
	92	0	X122 005
90	0		X122 010
	X32		X122 020
	X35		X122 030
	X37		X122 040
	91		X122 050
	X45		X122 060
	X170	0	X122 070
91	0		X122 080
	X151	0	X122 090
92	21(0)		X122 100
X123	90		X123 000
	92	0	X123 005
90	0		X123 010
	X32		X123 020
	X35		X123 030
	X37		X123 040
	91		X123 050
	X45		X123 060
	X170	0	X123 070
91	0		X123 080
	X151		X123 090
	X150		X123 100
	X150	0	X123 110

ARG. X124 ONO	92	21(2)		X123	120
	X124	90	0	X124	000
	90	0		X124	010
		X32		X124	020
		X36		X124	030
		X37		X124	040
		91		X124	050
		X45		X124	060
		92	0	X124	070
	91	0	0	X124	080
	92	93	0	X124	090
	93	0		X124	100
		X33		X124	110
		L14		X124	120
		X34		X124	130
		N0	0	X124	140
ARG. X125. VARIABLE S2.	X125	90	0	X125	000
	90	0		X125	010
		X32		X125	020
		X36		X125	030
		X37		X125	040
		91		X125	050
		X45		X125	060
		X4	0	X125	070
	91	0	0	X125	080
ARG. X126. VARIABLE S0.	X126	90	0	X126	000
	90	0		X126	010
		X32		X126	020
		X36		X126	030
		X37		X126	040
		91		X126	050
		X45		X126	060
		X3	0	X126	070
	91	0	0	X126	080
	X150	0		X150	000
		911		X150	010
		912		X150	020
		913	0	X150	030
	911	21THE N		X150	001
	912	21EXT O		X150	002
	913	21F		X150	
	X151	0		X151	000
		911		X151	010
		912		X151	020
		913	0	X151	030
	911	21THE C		X151	001
	912	21ONTEN		X151	002
	913	21TS OF		X151	003
SYMB H0	X170	90	0	X170	000
	90	0		X170	010
		X32		X170	015
		X35		X170	016
		X33		X170	020
		L8		X170	030

		X34		X170	040
		N0	0	X170	050
SYMB H1	X171	90	0	X171	000
	90	0		X171	010
		X32		X171	015
		X36		X171	016
		X33		X171	020
		L8		X171	030
		X34		X171	040
		N1	0	X171	050
SYMB H2	X172	90	0	X172	000
	90	0		X172	010
		X32		X172	015
		X36		X172	016
		X33		X172	020
		L8		X172	030
		X34		X172	040
		N2	0	X172	050
SYMB. H5	X175	90	0	X175	000
	90	0		X175	010
		X32		X175	015
		X36		X175	016
		X33		X175	020
		L8		X175	030
		X34		X175	040
		N5	0	X175	050
LIST OF SEGMENTS FOR	X180	0		X180	000
COMPOSING J77		90		X180	005
		91		X180	010
		92		X180	015
		93	0	X180	020
	90	0		X180	025
		94	0	X180	030
	94	940	0	X180	035
	940	0		X180	040
		X45		X180	045
		941	0	X180	050
	941	942	0	X180	055
	942	0		X180	060
		X33		X180	065
		L10		X180	070
		X34		X180	075
		N50	0	X180	080
	91	0		X180	085
		95	0	X180	090
	95	950	0	X180	095
	950	0		X180	100
		X45		X180	105
		951	0	X180	110
	951	952	0	X180	115
	952	0		X180	120
		X33		X180	125
		L10		X180	130
		X34		X180	135

	N60	0	X180	140
92	0		X180	145
	96		X180	150
	97		X180	155
	98	0	X180	160
96	960	0	X180	165
960	0		X180	170
	X43		X180	175
	N1		X180	180
	X44		X180	185
	N2		X180	190
	X45		X180	195
	961	0	X180	200
961	962	0	X180	205
962	0		X180	210
	X33		X180	215
	L8		X180	220
	X34		X180	225
	N0	0	X180	230
97	970	0	X180	235
970	0		X180	240
	X43		X180	245
	N1		X180	250
	X44		X180	255
	N1		X180	260
	X45		X180	265
	971	0	X180	270
971	972	0	X180	275
972	0		X180	280
	X33		X180	285
	L23		X180	290
	X34		X180	295
	N0	0	X180	300
98	980	0	X180	305
980	0		X180	310
	X45		X180	315
	981	0	X180	320
981	982	0	X180	325
982	0		X180	330
	X33		X180	335
	L10		X180	340
	X34		X180	345
	N2	0	X180	350
93	0		X180	355
	99		X180	360
	910	0	X180	365
99	990	0	X180	370
990	0		X180	375
	X43		X180	380
	N3		X180	385
	X44		X180	390
	N0		X180	395
	X45		X180	400
	991	0	X180	405

FLOW DIAGRAM FOR J77

991	992	0	X180	410
992	0		X180	415
	X33		X180	420
	L8		X180	425
	X34		X180	430
	N0	0	X180	435
910	995	0	X180	440
995	0		X180	445
	X45		X180	450
	996	0	X180	455
996	997	0	X180	460
997	0		X180	465
	X33		X180	470
	L10		X180	475
	X34		X180	480
	N30	0	X180	485
X181	0		X181	000
	90		X181	010
	91		X181	020
	92		X181	030
	93	0	X181	040
90	905	0	X181	050
905	0		X181	060
	X41		X181	070
	N1		X181	080
	X43		X181	090
	N0		X181	100
	X46		X181	110
	N2	0	X181	120
91	915	0	X181	130
915	0		X181	140
	X41		X181	150
	N2		X181	160
	X43		X181	170
	N7		X181	180
	X45		X181	190
	N4		X181	200
	X46		X181	210
	N3	0	X181	220
92	925	0	X181	230
925	0		X181	240
	X41		X181	250
	N3		X181	260
	X43		X181	270
	N7		X181	280
	X45		X181	290
	N2		X181	300
	X46		X181	310
	N4	0	X181	320
93	935	0	X181	330
935	0		X181	340
	X41		X181	350
	N4		X181	360
	X43		X181	370

		N0		X181 380
		X46		X181 390
		N0	0	X181 400
ROUTINE J77	X182	90	0	X182 000
	90	0		X182 010
		X41		X182 020
		91	0	X182 030
	91	92	0	X182 040
	92	0		X182 050
		X33		X182 055
		L10		X182 060
		X34		X182 070
		N77	0	X182 080
LIST OF COMPILED ROUTINES WITH DSCNS.	X196	0		X196 000
		X101		X196 010
		X102		X196 020
		X104	0	X196 030
	X197	0		X197 000
		X20		X197 010
		X24	0	X197 020
	X198	90	0	X198 000
	90	0		X198 010
		X20		X198 020
		U134		X198 030
		X24		X198 040
		U140	0	X198 050
DESCRIBABLE N-LIST	X199	0		X199 000
		N0		X199 010
		N1		X199 020
		N2		X199 030
		N3		X199 040
		N4		X199 050
		N5		X199 060
		N6		X199 070
		N7		X199 080
		N8		X199 090
		N9	0	X199 100
KICKOFF	5	T1		T1 000
				002793

Appendix B

PROGRAM LISTING OF AN INFORMATION-ANNEXING SCHEME

The listing in this Appendix is for an annexing routine, capable of annexing to memory the contents of sets of sentences of the kinds considered in Part III.

2 A		10	
2 M		100	
2 N		100	
2 X		200	
2 Y		200	
5			
5306M0. PREPARE LIST (0) FOR	M0	40H0	M000010
3073 M1. OUTPUT (0) IS VALUE		10X99	M000020
3075 (1) IS FUNCTION		J62	M000030
3076 X99= ... IS THE		J75	M000040
3077 VALUE OF ...		J50	M000050
3078PUT FUNCTION IN W0		40H0	M000060
3080		J70	M000070
3081		40H0	M000080
3083IF VALUE IS SYMBOL		J60	M000090
3084FIND IT, AND DO 90		70J7	M000095
3086		J60	M000100
3087		30H0	M000110
3091		70	90 M000120
3093		J60	M000130
3094		52H0	90 M000140
3089BRING IN FUNCTION	90	11W0	M000150
3097 AND EXIT		J6	J30 M000160
5307ASSIGN (0) AS VALUE OF THE (1)	M1	J42	M001010
3090		J21	M001020
3098		11W1	M001025
3100		40H0	
3102		M20	M001026
3103SIMPLIFY (0)		M21	M001027
3104		40H0	
3106		M20	M001028
3107		20W1	M001029
3109		J71	
3110		J71	
3111	91	11W1	M001030
3114		J83	M001040
3118		7090	M001050
3120		30H0	M001060
3122		J90	M001070
3123		J136	M001080
3124		60W2	M001090
3129		910	M001110
3130		11W1	M001140
3132		J60	M001150
3133		J68	M001160
3134		11W2	M001170
3136		20W0	91 M001180
3116	90	11W1	M001190
3139		J82	M001195
3140		910	M001210
3141		11W1	M001230
3143		J71	J32 M001240
3127ASSIGN 1W0	910	11W0	M001250
3145 AS VALUE OF J81(1W1) OF (0).		11W1	M001260
3147		J81	M001270
3148		40H0	M001280
3150IF J81(1W1)=X98		10X98	M001290
3152 PUT 1W0 AT END OF (0)		J2	M001300
3153		70J11	M001310

3155		30H0	J65	M001320
5308	M2	M10		M002010
3128		40H0		
3117		J150		
3115		J4		M002015
3158		1090		M002020
3160		J100		M002025
3161		J154		
3162		J155		
3163		10X105		M002026
3165		M4	M7	M002027
3112	90	M0		M002030
3166		M1		M002040
3167		J154		M002045
3168		J155		M002046
3169		10X105		M002050
3171		J150	J4	M002060
5309DELFTF X98 TO X97 OF (0)	M3	10X97		M003010
3172		10X98	M26	M003020
5310M4, CREATE DESCRIPTION	M4	J90		M004010
3174 OF EXPRESSION (0)		J90		M04 020
3175OUTPUT (0) IS DESCRIPTION		J52		M04 030
3176		11W1		M004032
3178		10X100		M004034
3180		J65		M004036
31811W0 IS OUTPUT LIST		11W1		M04 040
31831W1 IS GENERATOR LIST		11W2		M04 050
31851W2 IS EXPRESSION		J64		M04 060
3186		11W1		M04 070
3190		1090		M04 080
3192		J100		M04 090
3193		11W0	J32	M004100
31881W2 IS CURRENT EXPRESSION	90	40H0		M004 0110
3196		J152		
3197		60W2		
3199FIND ITS TYPE		M5		M04 120
3200ADD IT TO END OF VALUE LIST		40H0		M04 130
3202 FOR TYPE		11W0		M04 140
3204		J6		M04 150
3205		11W2		M04 160
3207		J6		M04 170
3208		J13		M04 180
3209		11W2		M04 190
3211PUT COMPONENTS ON GEN. LIST		11W1		M004200
3213		M6	J4	M004210
5311M5, FIND TYPE OF EXPRESSION (0)	M5	J0		M005010
3189		12H0		M05 015
3215OUTPUT (0) IS TYPE		J60		M05 020
3216		30H0		M05 030
3220		70	90	M005050
3222		J60		M05 050
3223		30H0		M05 060
3228		7091		M05 070
3230		10X85	0	M05 080
3218	90	30H0		M005090
3233		10X86	0	M005095
3226	91	10X45	0	M05 100
3237	92	30H0	91	M005110
5312M6, PUT COMPONENTS OF (1)	M6	J52		M006010
3238 OF TYPE (2)		11W2		M06 020

32270N GENERATOR LIST (0)		10X45		M06 030
3219NO OUTPUT		J2		M06 040
3240		70	J32	M06 050
3242		11W0		M006052
3244		10X100		M006054
3246		J62		M006056
3247		20W0		M006058
3249		11W2		M06 060
3251		10X85		M06 070
3253		J2		M06 080
3256		7090		M06 090
3258		11W1		M006141
3261	96	J60		M006142
3266		7092		M006143
3268		12H0		M006144
3270		11W0		M006145
3272		J6		M006146
3273		J63	96	M006147
3254	90	12W1		M06 150
3276	91	J60		M06 160
3278		7092		M06 170
3280		J60		M006175
3281		12H0		M06 180
3283		11W0		M06 190
3285		J6		M06 200
3286		J63	91	M006210
3264	92	30H0	J32	M06 230
5313M7. PRINT DESCRIPTION (0)	M7	J40		M007010
3277 SAVE IT IN W0		60W0		M007020
3265 PRINTS ITS LIST OF DLS		10X86		M002030
3255		910		M007040
3288		11W0		M007050
3290 PRINT ITS LIST OF LIST		10X85		M007060
3292		910		M007070
3293		11W0		M007080
3295 PRINT ITS LIST OF SYMBOLS		10X45		M007090
3297		910	J30	M007100
3262	910	40H0		M007110
3299 PRINT ATTRIBUTE		J152		M007120
3300		J10		M007130
3304		7090		M007140
3306 PRINT LIST		J151		M007150
3302 SPACE	90	J154	J155	M007160
5316M10. READ LIST OF STATEMENT	M10	J0		M10005
3303		J154		
3301		10W25		M10010
3307 AND STORE AS LIST OF		M11		M10020
3308 LISTS (0).		10W30		M10030
3310		M11		M10040
3311SET UP OUTPUT LIST		J90		M10050
3312		J50		M10060
3313READ NEXT LINE	90	J180		M10070
3318		7091		M10080
3320SET UP STATEMENT LIST		J90		M10090
3321		J136		M10100
3322SET POINTER TO PSJ 0		11W25		M10110
3324		J124		M10120
3325SET POINTER TO FIRST SUMB		J184		M10130
3326		7091		M10135
3328		30H0		M10140

3331	92	11W30		M10150
3334		J124		M10160
3335MEASURE SYMR		J183		M10170
3336		30H0		M10180
3338 AND SAVE IT		40H0		M10100
3340AND INPUT IT TO H0		J181		M10190
3341PUT IT AT END OF STATEMENT LIST		J65		M10200
3342		11W25		M10210
3344FIND NEXT SYMR		J184		M10220
3345		30H0		M10230
3350		7093	92	M10240
3316	91	30W25		M10250
3353		30W30		M10260
3355		11W0	J30	M10270
3348	93	11W0		M10280
3358		J6		M10290
3359		J65	90	M10300
5317M11. SET UP D.T. IN (0)	M11	J50		M11010
3349		J90		M11020
3347		J124		M11030
3332		41W0		M11040
3317		21W0	J30	M11050
5326M20. CREATE FIND LIST	M20	J50		M020010
3314 FROM DESCRIPTION LIST (0)		J90		M020020
3360 OUTPUT (0) IS FIND LIST,		40H0		M020030
3362		11W0		M020040
3366		1090		M020050
3368		J100		M020060
3369		30H0	J30	M020065
3364	90	J64		M020070
3371		40H0	0	M020080
5327M21. REDUCE FIND LIST (0)	M21	J41		M021010
3365 OUTPUT (0) IS REDUCED LIST		40H0		M021020
3374		J60		M021030
3375		60W0		M021040
3377 FIND NEXT ARGUMENT		52H0		M021050
3379	92	11W0		M021060
3382LOCATE NEXT DETERMINER		J60		M021070
3386 IF NONE, GO TO 90		7090		M021080
3388FIND DETERMINER		52H0		M021090
3390 SAVE IT		60W1		M021100
3392DETERMINE FUNCTION		J10		M021110
3396IF NO VALUE, GO TO 91		7091		M021120
3398DELETE ARGUMENT FROM LIST		11W0		M021130
3400		J68		M021140
3401REPLACE DETERMINER WITH VALUE		61W0	92	M021150
3384END OF LIST	90	30H0		M021160
3404		30H0	J31	M021170
3394PROCESS X97 OR QUIT	91	11W1		M021180
3407		10X97		M021190
3409		J2		M021200
3410		70J31		M021210
3412POINT TO PREDECESSOR OF X97		11W0		M021220
3414AND PROCESS IT		M22		M021230
3418		7093		M021240
3420		61W0		M021250
3425		94	92	M021254
3423	94	J0		M021255
3426		11W0		M021256
3428		M24		M021260

3429DELETE X98		11W0		M021270
3431		J60	J68	M021280
3416CREATE OBJECT FOR	93	40W2		M021300
3433SURDESCRIPTION		11W0		M021310
3435COPY PART DESCRIPTION		M23		M021320
3436		20W2		M021330
3438CREATE NEW OBJECT		J90		M021340
3439		J136		M021345
3440		40H0		M021350
3442		40H0		M021360
3444		11W2		M021370
3446		J6		M021380
3447PUT IT AT HEAD OF PART DESCRIPT		J64		M021390
3448		12W0		M021400
3450		J6		M021410
3451PUT IT AT END OF ARG LIST		J65		M021420
3452		11W2		M021430
3454DESCRIBE IT		M20		M021440
3455		M0		M021450
3456		M1		M021460
3457SURSTITUTE IT FOR ARG		21W0		M021470
3459		94	J32	M021480
5328M22. IDENTIFY LIST MEMBER	M22	J0		M022010
3424		12H0		M022015
3417 WITH ARG. LOCATION (0)		J50		M022020
3415 X97 AT N(0)		M23		M022030
3395IF IDENTIFIED, OUTPUT		40H0		M022040
3385(0) IS NAME OF MEMBER,		11W0		M022050
3461IF NOT IDENTIFIED, NO		1090		M022060
3463OUTPUT. FIND LIST NOT		J100		M022070
3464		J5		M022072
3465		70	J30	M022074
3467		30H0		M022076
3469		30H0	J30	M022078
3380MUTILATED	90	60W0		M022080
3472		M25		M022090
3476		70	91	M022100
3478		40H0	J4	M022110
3474	91	51W0	J3	M022120
5329M23. COPY DESCRIPTION FROM	M23	J0		M023010
3475 X97 TO X98 OF (0)	91	J60		M023020
3460 OUTPUT (0) IS COPY		70J7		M023030
3482		12H0		M023040
3484		10X97		M023050
3486 TEST IF = X97		J2		M023060
3487		7091		M023070
3489 PUT LOC OF X97 OUT		J50		M023080
3490 CREATE NAME OF COPY		J90		M023090
3491		40H0		M023100
3493		11W0		M023110
3498DO 90 ON CELLS OF (0)		1090		M023120
3500		J100	J30	M023130
3496	90	40H0		M023140
3502IF SYMB IS X98, QUIT		10X98		M023150
3504		J2		M023160
3505		J5		M023170
3509		70	92	M023180
3511		J8	J8	M023190
3507IF NOT, ADD TO LIST	92	J65		M023200
3512		40H0	0	M023210

5330DELETE X97 TO X98 OF (0)	M24	10X98		M024010
3506		10X97	M26	M024020
5331M25. TEST IF (0) FITS	M25	J51		M025010
3495 DESCRIPTION (1)	91	11W0		M025020
3515		11W1		M025030
3517LOCATE NEXT DETERMINER		J60		M025040
3518SAVRE LOCATION		60W1		M025050
3520FIND IT		52H0		M025060
3522		40H0		M025070
3524		10X99		M025080
3526IS IT X99		J2		M025090
3530IF SO, TERMINATE TEST		70	90	M025100
3532FIND VALUE OF (0)		J10		M025110
3533IF NONE, EXIT WITH -		70J31		M025120
3535IF SO, SUBSTITUTE IT FOR (0)		20W0	91	M025130
3528COMPARE VALUE WITH VALUE	90	30H0		M025140
3538OF DESCRIPTION		11W1		M025150
3540		J60		M025160
3541		70J7		M025170
3543		52H0		M025180
3545AND EXIT WITH SIGNAL		J2	J31	M025190
5332M26 DELETE SECT FROM (0) TO (1)	M26	J52		M026010
3529OF LIT (0). NO OUTPUT.		11W2		M026020
3473TESTS THAT CN(2)=(0). IF (1)		11W1		M026030
3547NOT ON LIST, DOES NOTHING		J77		M026040
3548		70J32		M026050
3550		11W2		M026060
3552		J60		M026070
3553		12H0		M026080
3555		11W0		M026090
3557		J2		M026100
3558		70J7		M026110
3560	90	J68		M026120
3562		11W2		M026130
3564		J60		M026140
3565		70J7		M026150
3567		12H0		M026160
3569		11W1		M026170
3571		J2		M026180
3572		7090		M026190
3574		30H0	J32	M026200
	5	1		
5619	X113			X113010
5296	A0	0		A000010
3561		X113		A000013
3576		X99		A000015
3577		X31		A000020
3578		X20		A000030
3579		X105	0	A000040
5297	A1	0		A001010
3580		X114		A001013
3581		X99		A001015
3582		X33		A001020
3583		X25		A001030
3584		X105	0	A001040
5298	A2	0		A002010
3585		X115		A002013
3586		X116		A002015
3587		X99		A002017
3588		X34		A002020

3589			X25		A002030
3590			X105	0	A002040
5407	N1	1			1 N001010
5530SDSC OF ROUTINE	X24		90	0	X24 010
3591	90		0		X24 020
3593			X32		X24 030
3594			X86		X24 040
3595			X87		X24 050
3599			91	0	X24 060
3597	91		0		X24 070
3600			X71	0	X24 080
5547NAME	X41		90	0	X41 010
3598	90		0		X41 020
3592			X32		X41 030
3601			X45	0	X41 040
5577LIST OF AFFECTED CELLS	X71		90	0	X71 010
3596	90		0		X71 020
3603			X32		X71 030
3604			X85		X71 040
3605			X91		X71 050
3606			X72	0	X71 060
5578AFFECTED CELL	X72		90	0	X72 010
3602	90		0		X72 020
3608			X32		X72 030
3609			X86		X72 045
3610			X87		X72 050
3614			91	0	X72 060
3612	91		0		X72 070
3615			X41		X72 080
3616			X75		X72 090
3617			X76	0	X72 100
5581INPUT STATE OF CELL	X75		90	0	X75 010
3613	90		0		X75 020
3607			X32		X75 030
3618			X85		X75 040
3619			X91		X75 050
3620			X45	0	X75 060
5582OUTPUT STATE OF CELL	X76		90	0	X76 010
3611	90		0		X76 020
3622			X32		X76 030
3623			X85		X76 040
3624			X91		X76 050
3625			X45	0	X76 060
5591LIST		X85			X85
5592DESCRIPTION LIST		X86			X86
5593ATTRIRUTE		X87			X87
5594OBJECT		X88			X88
5595VALUE		X89			X89
5597MEMBER TYPE		X91			X91 010
112	5		M2		
3630	3	0			
	2	3627			
	2	3637			
	2	3644			
3627	3	0			
	0	X113			
	0	X99			
	0	X31			
	0	X20			